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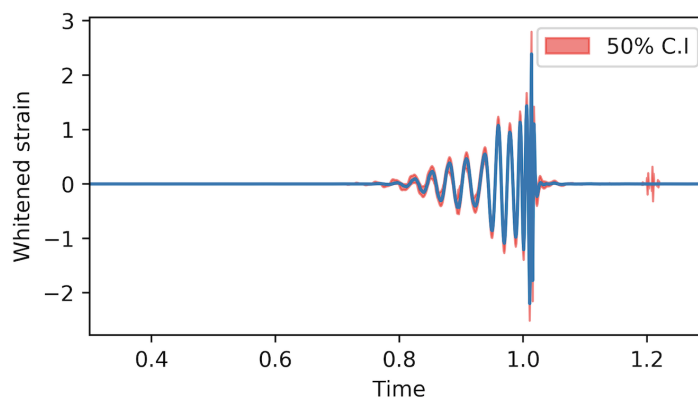
Gravitational-wave burst searches with multi-detector networks

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Statement of Contribution

- Chapter 1 is an original review of the literature relevant to this research. Figures presented in this chapter are results from previous work and are cited accordingly.
- Chapter 2 provides a summary of the *BayesWave* algorithm based on the original literature by N. Cornish and T. Littenberg [22], and also an extension to the analytic results presented in [45]. Unless otherwise noted, the calculations in Section 2.3.3 are my own.
- Some of the language in the abstract and Chapters 1-2 is taken from the paper listed below.
- Chapter 3 discusses the methods of obtaining data for this work. Sources are cited accordingly.
- Chapter 4 shows the results of preliminary tests conducted to assess *BayesWave*'s performance in generic signal and glitch identification. Results presented in this chapter are my own.
- Chapter 5 presents an extension of the results in a paper in the final stages of preparation authored by myself, Dr. Margaret Millhouse and Prof. Andrew Melatos

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- Results in Chapters 4 and 5 are obtained using the *BayesWave* pipeline as cited in Chapter 2.
- All figures from Chapter 3 - 5 are my own. I drafted the text, with substantial edits from Dr. Margaret Millhouse and Prof. Andrew Melatos.

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THE UNIVERSITY OF MELBOURNE

Abstract

Master of Science (Physics)

Gravitational-wave burst searches

with multi-detector networks

by Yi Shuen Christine LEE

In 2015 the advanced Laser Interferometer Gravitational-Wave Observatory (LIGO) began its first Observing Run. LIGO was joined by Virgo in 2017, and the Kamioka Gravitational-Wave Detector (KAGRA) in 2020. The increasing number of detectors, $\mathcal{I}$ in the global gravitational-wave detector network will lead to higher rates of detection, better parameter estimates, and more precise sky localisation to aid in electromagnetic follow-up. In this work, we investigate the impact of expanding detector networks in gravitational-wave burst detection and reconstruction using the *BayesWave* algorithm. *BayesWave* is a source-agnostic analysis that reconstructs features in detector data with multiple models, and distinguishes between true astrophysical signals and instrumental glitches.

The Bayes factor between the signal and glitch model, $\mathcal{B}_{S,G}$ is the figure of merit for detection confidence. We first derive an analytic scaling of $\ln \mathcal{B}_{S,G}$ with signal strength, waveform complexity and $\mathcal{I}$. We then test for the consistency of *BayesWave* by using it to recover injected simple sine-Gaussian signal waveforms and coincident glitches from simulated detector noise. The sine-Gaussian signal recoveries preliminarily show a good agreement between analytic and empirical results. The glitch injection recoveries show that *BayesWave* is able to correctly identify glitches that can potentially resemble astrophysical waveforms. Upon testing the functionality of *BayesWave*, we inject 150 phenomenological binary black hole (BBH) waveforms with equal component masses of $30M_{\odot}$ into simulated Gaussian noise of Hanford-Livingston (HL), Hanford-Livingston-Virgo (HLV) and Hanford-Livingston-KAGRA-Virgo (HLKV) networks at projected sensitivities for the fourth observing run, O4. Our empirical results show consistency with the analytic results such that *BayesWave* is able to recover signals with higher Bayes factor and hence higher confidence with increasing $\mathcal{I}$. We also compare the network overlap between injected and recovered signals, and find that increasing $\mathcal{I}$ also allow for more accurate waveform reconstructions. In addition, we quantify the accuracy of *BayesWave* in sky localisation. We observe significant reduction in search area, $\mathcal{A}$ and sky areas enclosed within 50% and 90% credible intervals with increasing $\mathcal{I}$ indicating that sky localisation improves with expanded detector networks. Finally, we compare the accuracy of *BayesWave* in estimating model-independent waveform moments which include the central time, signal duration, central frequency and bandwidth. Although we did not observe any improvements in waveform moment estimations with expanding detector network, we show that the accuracy improves with signal strength and that higher-order waveform moments are estimated with larger error bars.

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Chapter 1

Introduction

Gravitational waves are gravitational fluctuations propagating across the fabric of space-time as a consequence of accelerating massive bodies. This phenomenon was postulated by Albert Einstein in 1915 and exactly 100 years later in September 2015, the two detectors of the Laser Interferometer Gravitational-Wave Observatory (LIGO) at Hanford and Livingston detected the very first gravitational-wave signal (GW150914) in history during their first observing run, O1. This signal originated from a merger between two black holes at a luminosity distance of 410^{+160}_{-180} Mpc, with initial masses of $36^{+5}_{-4}M_{\odot}$ and $29^{+4}_{-4}M_{\odot}$ [3]. A Nobel prize was awarded for this ground-breaking discovery in 2017 for it has opened up countless windows of opportunities for new research in astronomy.

For many years, scientists have been exploiting electromagnetic waves emitted by astrophysical sources to image and understand astrophysical processes that are happening in the vast universe. One of many challenges in traditional astronomy, which observes using electromagnetic radiation, is that telescopes must be able to capture sufficient photons emitted by the target source to obtain decent quality spectra or images for analysis. This makes it is extremely difficult to image and study faraway sources given that photons propagating from such distances will encounter numerous obstructions throughout their journey towards earth. By exploiting the unobstructed propagation of gravitational waves, astronomers are now able to explore a new regime of space previously inaccessible by electromagnetic radiation and retrieve information on obscure astrophysical sources, like black holes which do not emit photons. This can lead to new and profound discoveries.

Between 2015 and 2020, the Advanced LIGO detectors [1, 13] in Hanford (H) and Livingston (L) have completed three observing runs: O1 [4, 6], O2 [5, 6] and O3 [31], and in addition the conducted joint searches with Italian partner Virgo [10] in the late stages of O2 and the whole of O3. The Kamioka Gravitational-Wave Detector (KAGRA) [11, 12, 14] also began observing in February 2020, towards the end of O3. Over the duration of O1, the advanced gravitational-wave detector network has confirmed detections of three binary black hole (BBH) mergers and in O2, seven BBH mergers and one binary neutron star (BNS) merger [6]. Furthermore, there are 56 candidates for gravitational-wave events that are yet to be confirmed from O3 [31]. This is a remarkable milestone, and it is definitely a promising start to an exciting era of gravitational-wave astronomy.

1.1 Detecting Gravitational Waves

1.1.1 Gravitational Waves

The existence of gravitational waves is postulated by Einstein's Theory of General Relativity. The linearised Einstein's field equations in natural units ($c = G = 1$) is given by

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu} \quad (1.1)$$

assumes that the gravitational field metric is given by $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon h_{\mu\nu}$, where $\eta_{\mu\nu}$ is the Minkowski (flat) metric and $h_{\mu\nu}$ is a perturbation term. Linearisation in this context arises from the fact that $0 < \epsilon \ll 1$ such that all perturbation terms with ϵ^k with $k > 1$ can be neglected.

This linearised theory then leads to the weak-field Einstein equations

$$\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu} \quad (1.2)$$

where $\square = \eta_{\mu\nu} \partial^\mu \partial^\nu$ is the D'Alembertian (wave) operator and $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h_{\alpha\beta} \eta^{\alpha\beta}$. In vacuum, the metric tensor, $T_{\mu\nu}$ is zero. When inside the source of the field, $T_{\mu\nu}$ is non-zero and the solution, $h_{\mu\nu}$ is a plane wave travelling at the speed of light, which is what we call gravitational waves [34, 58].

1.1.2 LIGO Data

Gravitational-wave detectors in the global network are large-scale Fabry-Pérot interferometers. The two LIGO detectors H and L each have arms of length 4 km, where as the Virgo (V) and KAGRA (K) detectors each have two arms of length 3 km. These detectors are sensitive to minuscule deformation in space-time caused by the propagation of gravitational waves. With equal arm lengths, light destructively interferes at the output sensor. In presence of gravitational waves, the change in arm length causes the light beams from both arms to no longer interfere perfectly at the output sensor. The interference pattern of light from the two beams encodes information on space-time deformation.

The degree of space-time deformation is quantified by *strain*, a dimensionless parameter defined as the fractional change in arm lengths of each detector

$$\vec{h}(t) = \frac{\Delta L(t)}{L} \quad (1.3)$$

where ΔL is the change in arm length and L is the original arm length. In other words, strain is the amplitude of gravitational-wave signals. The vector $\vec{h}$ is an array of strain for a range of discrete timesteps and $\vec{h}(t)$ is the strain at a specific timestep, t . In the case of the first detection, GW150914, the peak strain was 1.0×10^{-21} , a remarkably small change in length [3]. This underlines the level of precision that gravitational-wave detectors have to achieve in order to make significant detections.

With strain sensitivities of up to 10^{-23} [48], gravitational-wave detectors are prone to various sources of noise, notably the displacement noises induced by thermal fluctuations and seismic motion, as well as shot noise due to quantum fluctuations in the local light field [61]. On top of that, results from the initial LIGO-Virgo science runs indicated non-stationary and non-Gaussian detector noises, which includes short-duration noise transients called 'glitches'. Glitches can be of environmental origin or due to instrumental sources like the Voltage controlled oscillators [52, 66], but in

many cases the source of the glitch is unknown [8]. From time to time, noise artefacts like glitches can resemble gravitational waves and consequently limit the ability of detecting low-amplitude signals. This raises a challenge for the search of astrophysical signals embedded within detector data.

Generally, the dimensionless strain of detector data, $\vec{s}_i$ for the i -th detector in the network can be written in terms of the noise component $\vec{n}_i$ and a potential signal waveform, $\vec{h}_i(\vec{\theta})$ which should be coherent across all detectors, and depends on some parameter vector $\vec{\theta}$ such that [45]

$$\vec{s}_i = \vec{h}_i(\vec{\theta}) + \vec{n}_i. \quad (1.4)$$

The noise component can be broken down further into $\vec{n} = \vec{n}_G + \vec{g}$ with $\vec{n}_G$ denoting the stationary (i.e. constant with time) coloured Gaussian noise component and $\vec{g}$ denoting a potential instrumental glitch that may be present in the data [22]. In signal processing, coloured noise refers to non-uniform power spectrum where the noise have higher power in certain frequency bands. White noise, on the other hand, has a uniform power spectral density across all frequencies in a given range.

The big question in gravitational-wave data analysis is: Does the detector data contain a real, astrophysical signal? If there is, what is its morphology? The first step is to understand the noise properties of the initialised detector data in absence of a signal and constructing noise models comparable to real detector noise. Noise models can then be subtracted from detector data, $\vec{s}$, to acquire the signal component as per Equation 1.4. This can be easily achieved when the detector noise is stationary and Gaussian, but when these conditions do not hold, an alternative approach is required. This will be discussed further upon the introduction of a detection algorithm called *BayesWave*.

1.2 Gravitational-wave Bursts

Gravitational-wave signals can be divided up into two subgroups: (i) transient signals (bursts) which have durations ranging from a few milliseconds up to a few seconds; (ii) long duration signals which can last up to hundreds of seconds or more. Since gravitational-wave signals originate from a large variety of sources, each of the two groups can be further divided into groups of signals with sources that have been well-modelled, and sources that remain relatively unmodelled.

BBH mergers are the most common burst sources, and are more generally classified as compact binary coalescences (CBC), a general term used to describe a system of two inspiralling and merging compact objects like neutron stars and black holes. CBCs are the most well-understood gravitational-wave source to date and there is an extensive collection of well-developed models available to predict the waveforms emitted from such systems. For that reason, searches for CBCs can be done using a technique called matched-filtering, which will be discussed in more detail later in Section 1.3.

Although CBC signals are the only type of gravitational-wave signal with confirmed detection so far, there are many other transient astrophysical processes like core-collapse supernovae (SNe) of massive stars [28], pulsar glitches [62] and cusps in cosmic strings [25] which can be potential candidates for future burst signal detection. Due to the complexity and non-deterministic nature of some of these sources, there are not robust waveform templates available for these signals like there are for CBC signals. This makes it difficult to identify unfamiliar astrophysical signals when buried amongst noise in detector data. Moreover, unmodelled signals from unmodelled sources

could be more easily misidentified as glitches. Despite the challenges, it is worthwhile to search for generic gravitational-wave burst signals as they can provide remarkable insights to the physics of the originating source: from the dynamics of its surrounding environments to the centre of the object itself. This can improve our understanding of obscure astrophysical sources and processes, or even make prospective discoveries of sources previously unknown to humankind.

Long duration gravitational waves, on the other hand, have been predicted to exist by general relativity but no detection has been made for such signals just yet. Some potential sources for long duration gravitational include continuous waves from non-symmetric spinning neutron stars [60, 63] and stochastic gravitational-wave backgrounds [20] on a cosmological scale. Research in this area is a work in progress in both theoretical and technical aspects.

1.3 Matched-filtering Burst Searches

For both gravitational-wave detection and parameter estimation, we need to define the likelihood that the gravitational-wave strain data, $\vec{s}$, contains a signal, $\vec{h}$. Assuming stationary, uncorrelated Gaussian noise between all $\mathcal{I}$ detectors in the interferometer network, the likelihood of measuring data, $\vec{s}$ given gravitational-wave signal, $\vec{h}$ with parameters $\vec{\theta}$ is [8]:

$$p(\vec{s} | \vec{\theta}) = C \exp \left[-\frac{1}{2} \sum_{i=1}^{\mathcal{I}} \left(\vec{s}^i - \vec{h}^i(\vec{\theta}) | \vec{s}^i - \vec{h}^i(\vec{\theta}) \right) \right]. \quad (1.5)$$

where C is a constant. $(\cdot | \cdot)$ is the noise-weighted inner product. We define the noise-weighted inner product between two arbitrary waveforms $h_a(t)$ and $h_b(t)$ as [37]

$$(h_a | h_b) = 2 \int_0^\infty \frac{\tilde{h}_a^*(f) \tilde{h}_b(f) + \tilde{h}_a(f) \tilde{h}_b^*(f)}{S_n(f)} df. \quad (1.6)$$

$S_n(f)$ is the one-sided power spectral density (PSD) of stationary, Gaussian noise such that

$$\langle \tilde{n}(f) \tilde{n}^*(f') \rangle = \frac{1}{2} \delta(f - f') S_n(f) \quad (1.7)$$

where $\langle \tilde{n}(f) \tilde{n}^*(f') \rangle$ is the ensemble average over a large set of noise realisations, with $\tilde{n}(f)$ denoting the Fourier transform (FFT) of the noise component in Equation 1.4. The asterisk $*$ denotes the complex conjugate [51]. Under the assumption that physical gravitational-wave signals and detector outputs are real, we note that $\tilde{n}^*(f) = \tilde{n}(-f)$ and $\tilde{h}^*(f) = \tilde{h}(-f)$, so it follows from Equation 1.7 that $S_n(f) = S_n(-f)$ [51]. Hence the inner product was computed by integrating over positive frequencies only instead of using the full Fourier integral and also why $S_n(f)$ was named the *one-sided* PSD. Since the likelihood function in Equation 1.5 depends on the noise-weighted inner product of residuals, $\vec{r}^i = \vec{s}^i - \vec{h}^i(\vec{\theta})$, the likelihood quantifies the difference between the data and the signal.

Searching for well-modelled gravitational-wave bursts like CBCs uses a signal processing technique called matched-filtering which uses a template bank for the potential gravitational-wave signal, $\vec{h}$. The templates, $\vec{h}_s$, cover a range of astrophysical parameters like mass ratio and spin values. Templates for CBCs contains three major temporal phases: (i) inspiral, (ii) merger and (iii) post-merger ringdown. The templates are constructed to match phenomenological observations using a

combination of post-Newtonian theory, the effective-one-body formalism and numerical relativity (see Section 3.4) [29]. If the model $\vec{h}_s$ is consistent with true waveform $\vec{h}$, then $\vec{r}$ is a good representation of instrumental noise $\vec{n}$. Unmodelled burst searches, on the other hand, have no *a priori* model for the signal waveform. Therefore we must use alternative techniques to derive $\vec{h}_s$ directly from the data itself. In Section 1.4.1, we introduce one particular method for such an unmodelled search.

1.4 Pipelines for Gravitational-Wave Searches

Match-filter searches were performed on O1 and O2 detector data using three separate pipelines namely PyCBC [65], GstLAL [49, 57] and oLIB [46], which resulted in multiple high-significance detections of CBC signals. SPIIR is also a match-filter algorithm for CBC signal searches, but it is designed to make fast follow-up on electromagnetic events which might be related to a low-latency gravitational-wave triggers. Nevertheless, matched-filtering is only effective for characterising well-modelled signal waveforms and is not applicable to unmodelled, generic bursts like those mentioned in Section 1.2.

The alternative to matched-filter searches are unmodelled searches. The distinctive feature between the two methods is that unmodelled searches reconstructs waveform from detector data from scratch instead of fitting detector data using pre-determined templates. Apart from that, both types of searches use similar techniques and formalisms for model selection. To test for consistency between different search pipelines, unmodelled searches using coherent WaveBurst [39] (cWB) and *BayesWave* [22] were conducted as follow-up searches for gravitational-wave events [6] of O1 and O2. In fact, *BayesWave* is used as a follow-up to cWB and it helps increase confidence in burst detections [45]. Table 1.1 compares the various pipelines available for gravitational-wave burst searches.

Table 1.1: Summary table for comparing gravitational-wave search pipelines.

Detection pipeline	Waveform analysis	Templates	Search targets	Detection statistic
PyCBC [65]	Matched-filter	CBC waveforms	CBCs	Quadrature sum of reweighted SNR
GSTLal [57]	Matched-filter	CBC waveforms	CBCs	Multi-dimensional likelihood ratio
oLIB [46]	Matched-filter	sine-Gaussian templates	Generic burst	Likelihood ratio
SPIIR [35]	Time domain matched-filter	Infinite impulse response (IIR) filters	CBCs	Filter coefficient
cWB [40, 41]	Unmodelled	Waveforms reconstruction using solutions of likelihood functionals	CBCs, generic bursts	Maximum-likelihood ratio
BayesWave [22]	Unmodelled	Waveforms reconstruction by summing sine Gaussian wavelets	Generic bursts	Bayes factor

1.4.1 *BayesWave*

In this work, we focus on the *BayesWave* algorithm which uses a Bayesian approach for joint detection and characterisation of generic gravitational-wave bursts and glitches [22]. This algorithm was used to perform follow-up analyses with minimal assumption for GW150914 [2] and all-sky transient searches in O1 and O2 [4, 5]. The name *BayesWave* manifests the two key concepts adopted by the pipeline: (i) waveform reconstruction using sine-Gaussian wavelets and (ii) Bayesian model selection. In principle, *BayesWave* reconstructs transient features in a stretch of detector data, regardless of its origin, as three separate models: Gaussian noise only, Gaussian noise with instrumental glitches and Gaussian noise with a gravitational-wave signal. Each model is constructed by summing up a set of sine-Gaussian wavelets, with their parameters drawn using a trans-dimensional Markov Chain Monte Carlo algorithm. *BayesWave* then evaluates the evidences of all three models to assess their compatibilities with the data via the Bayes factor. This model selection technique is known as the Bayesian Model Selection. A detailed overview of the algorithm is presented in Chapter 2.

1.5 Benefits of Expanding Detector Networks

Increasing the number of operational ground-based detectors has several major benefits for gravitational wave astronomy, including a higher rate of detection of gravitational-wave transients, and better characterisation of those signals. Here we discuss some of the benefits of adding new detectors to the existing network.

1.5.1 Network SNR

Larger detector networks can more confidently detect quieter events. The strain amplitude, $s_i(t)$ of detector i of the network consists of a signal, h_i (if present) and detector noise, n_i as indicated by Equation 1.4. The squared matched-filter SNR of signal h_i in detector i is then given by [23]

$$\text{SNR}_i^2 = \left(\vec{h}_i \middle| \vec{h}_i \right) \quad (1.8)$$

where the right-hand-side of the expression is the noise-weighted inner product between the signal waveforms as per Equation 1.6. We then obtain the network SNR (SNR_{net}) of a gravitational-wave signal by summing the inner products from each interferometer such that

$$\text{SNR}_{\text{net}}^2 = \sum_{i=1}^{\mathcal{I}} \text{SNR}_i^2 \quad (1.9)$$

where $\mathcal{I}$ denotes the number of detectors in the network [45]. According to Equation 1.9, adding more detectors to the network increases the SNR of all detected gravitational-wave signal. High SNRs enables detection pipelines to detect gravitational-wave events with higher confidence and make more accurate estimates on the parameters of the waveform present in detector data [30].

In addition, the SNR of gravitational-wave signals scales with luminosity distance, D_L as [19]

$$\text{SNR}_i \propto \frac{1}{D_L}. \quad (1.10)$$

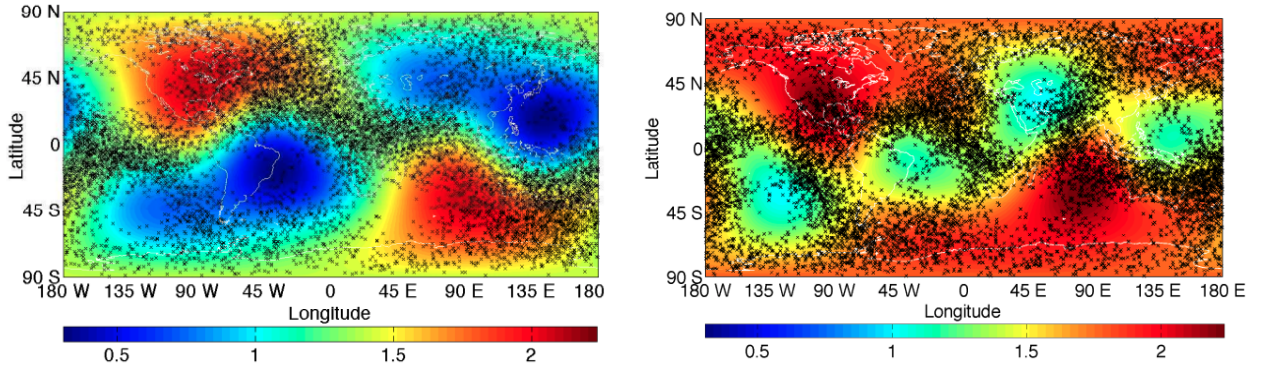


Figure 1.1: Network antenna power pattern over the sky for three (left panel) and four (right panel) detector network as in Figures 2 and 3 of [47]. Each black cross represent 10000 detected sources and the colour scale indicates the magnitude of network antenna power pattern.

By combining Equations 1.9 and 1.10 and assuming coherent searches, the overall SNR for a network of $\mathcal{I}$ detectors of equal sensitivities is given by $\text{SNR}_{\text{net}} \propto \sqrt{\mathcal{I}}/D_L$. In other words, an $\mathcal{I}$ -detector network can detect sources that are $\sqrt{\mathcal{I}}$ times further compared to a single detector. Assuming that gravitational-wave sources are uniformly distributed across the sky, an $\mathcal{I}$ -detector network can detect up to $\sqrt{\mathcal{I}^3}$ more sources compared to a single detector since the search volume scales as $V \propto D_L^3$.

1.5.2 Sky Coverage

The antenna power pattern of a detector network defines a region of space around the earth within which a source can be detected, up to a radial scale [59]. Hence the antenna power pattern affects the detection volume and sky coverage. In Reference [59], sky coverage is defined as the fraction of the sky over which the total network antenna power pattern is greater than half of its maximum value. The antenna power pattern of a detector at a particular sky location determines its directional sensitivity. Detectors in a global network are at a different geographical locations and are aligned differently, so the detectors are not equally sensitive to every sky location i.e. the detectors have different antenna patterns at different sky locations. Regions in the sky with more antenna pattern overlap between the detectors have higher total antenna patterns, thereby extending the search volume within the solid angles of these regions. Adding more detectors to the network at different orientations and geographical locations will introduce more antenna pattern overlaps, hence improving the sensitivity of the network to a wider region of the sky and consequently increase the detection volume along those directions [59].

Figure 1.1 shows a comparison between the network antenna pattern across the whole sky between a three-detector (HLV) network and a four-detector (HLKV) network. As expected, results show that both networks are more sensitive to some regions in the sky than others. However, the HLKV network has higher overall network antenna power pattern and an overall increase in sky coverage is also reflected in the expansion of regions with relatively higher sensitivity. Nevertheless, networks with antenna patterns of minimal overlap can be useful in performing all-sky surveys. Even though detection volume will be reduced since the total antenna power pattern is distributed across a large area, the network will have a more uniform (isotropic) sky coverage.

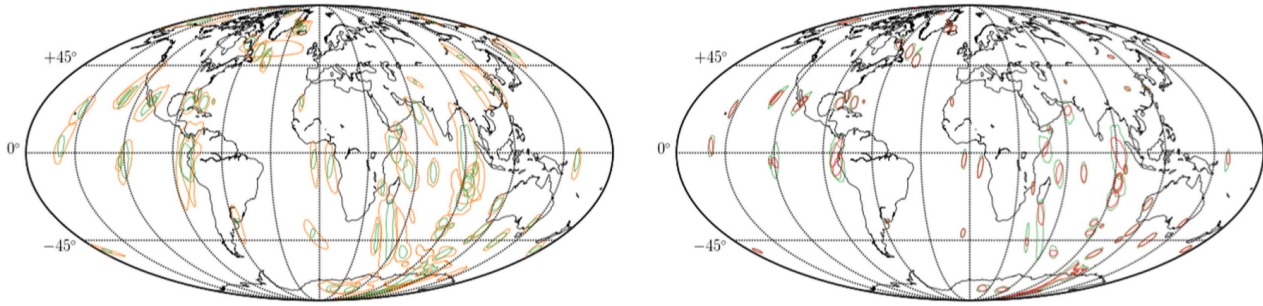


Figure 1.2: The left panel is a skymap of 90% credible regions for sky localisation of compact objects with the HLV (orange) and HLKV (green). Similarly in the right panel, for HLKV (green) and HLKIV (red). These skymaps originally appear in Figure 1 of [54].

1.5.3 Detection Rate

We also expect larger detector networks to have higher rates of detection due to extended network duty cycles. Duty cycle is the fraction of time a detector spends in its stable observing mode throughout the observation run. In order to enable a high confidence detection, the signal must be observed by more than one detector at different geographical locations and they must be coherent. Having a larger detector network increases the amount of time where two or more detectors are functional and simultaneously observing which consequently increases the chances of the detectors picking up a coherent astrophysical signal [59]. Increasing duty cycles with expanded detector networks was evident during O2. The combined (network) duty factor of the two LIGO detectors is 45%. With Virgo joining the network, the observing run operated with a duty factor of about 80% [5, 6].

1.5.4 Sky Localisation

When a gravitational-wave signal is identified, it is crucial to locate the origin of the signal, especially when the gravitational-wave event may be accompanied by electromagnetic counterparts. In the case of GW170817, the first ever detection of a binary neutron star (BNS) merger, sky localisation of the gravitational-wave source was of vital importance for locating and identifying electromagnetic counterparts to the gravitational-wave event [54]. Ground-based gravitational-wave detectors are nearly omnidirectional, so with a single detector we are not able to impose a strict constraint to the sky location of a gravitational-wave event. Sky localisation of gravitational-wave signals improves significantly with multiple interferometers. The difference in time of arrival between any two detectors will constrain the position of the source to an error ellipse in the sky map. While a two-detector network can roughly identify the sky location of an event to within an ellipse, adding extra detectors narrow down the size of the ellipse much further. For example: in a three-detector network consisting of detectors A , B and C , there will be three overlapping ellipses, each of which comes from the difference in time of arrival for detector pairs AB , AC and BC . These ellipses represent regions in the sky with similar arrival times to the values measured. Regions with more intersecting error ellipses indicate higher consistency of the source location with the measured arrival time differences. Therefore, having more detectors reduces localisation volume by imposing stricter constraints to the location of the source [26, 54]. Figure 1.2 shows two comparisons of 90% credible regions for compact object sky localisation: (a) between the HLV and HLKV networks on the left panel and (b) between

the the HLKV and HLKIV networks on the right panel, where “I” denotes the LIGO detector in India which is expected to operate in the near future. The reduction in 90% credible regions of the same event with increasing number of detectors suggests a universal improvement in sky localisation with expanded detector networks.

In sum, the advantages of having more detectors in the network include: (i) improvement in SNR and increased search volume, (ii) alignment-dependent sky coverage, (iii) increased rates of detection, and (iv) reduced sky localisation.

1.6 Motivation and Outline

Littenberg *et al.* [45] previously presented an analysis on the data of a two-detector network, consisting of the LIGO Hanford (H) and Livingston (L) detectors, where simulated gravitational-wave signals were injected into detector noise and recovered using *BayesWave*. The objective of this study is to evaluate the performance of *BayesWave* in identifying gravitational-wave transients amongst noise transients by means of the Bayes factor. In this thesis, we aim to extend the performance evaluation of *BayesWave* beyond the two-detector network by quantifying and comparing the performance of the *BayesWave* algorithm in gravitational-wave burst detection between detector networks of different sizes (i.e. different number of interferometers), using Bayes factor as the figure of merit. The general strategy is to inject the same set of simulated CBC waveforms into simulated detector data for networks of different sizes, recover the waveforms respectively from each network using *BayesWave* and then compare the empirical Bayes factors between them.

We establish the framework of this study in Chapter 2 by providing a detailed overview of the *BayesWave* algorithm, along with a derivation for the theoretical scaling of Bayes factor with number of detectors. In Chapter 3, we discuss the properties of simulated detector noise as well as the three types of waveforms used in this study. We also discuss the methods of obtaining the injected waveforms. For the purpose of assessing the performance of *BayesWave* in signal identification, we first use the algorithm to recover sine-Gaussian wavelets from simulated, stationary and Gaussian detector noise. Even though adding more detectors into the network have several advantages, it inevitably introduces more non-Gaussian and non-stationary features like glitches into detector data. This further challenges gravitational-wave searches, especially unmodelled searches where we have no prior knowledge of the waveform. Hence it is also non-trivial to test the algorithm’s ability in identifying transient noise features like glitches amongst Gaussian noise and distinguishing them from real gravitational signals. For this reason, we put forward a second consistency test by injecting simulated glitches into Gaussian detector noise for networks of different sizes and recovering them using *BayesWave*. We present the results of these preliminary tests in Chapter 4.

Upon testing for consistency and functionality of the algorithm, we inject simulated binary black hole signals into simulated noise of the HL, HLV and HLKV networks to evaluate *BayesWave*’s performance in gravitational-wave burst detections with detector networks of different sizes. We present the outcomes in Chapter 5 by comparing four aspects of waveform recovery: (i) Bayes factor, (ii) overlap between injected and recovered waveforms, (iii) sky localisation and (iv) parameter estimation. We conclude this research in Chapter 6 with a brief outlook for future work.

Chapter 2

The *BayesWave* Algorithm

This chapter overviews the *BayesWave* algorithm for the joint detection and characterisation of gravitational wave bursts and instrumental glitches. In Section 2.1, we discuss the underlying mathematical structure of sine-Gaussian wavelets and how sets of wavelets are parametrised for different reconstructed models. Section 2.2 discusses *BayesWave* obtains posterior distributions for model parameters. Section 2.3 presents a descriptive interpretation of the Bayes factor and is broken down into three subsections. We first define the Bayes factor and its application in model selection Section 2.3.1, followed by Section 2.3.2 where we mathematically justify how Occams Razor falls out naturally from the implication of Bayes' Theorem, we then derive an analytic scaling relation of Bayes factor with the number of detectors in the network in Section 2.3.3. Finally in Section 2.4, we introduce the network overlap which quantifies the overall match between reconstructed and injected waveforms across the detector network.

2.1 Sine-Gaussian Wavelets

BayesWave uses a sum of sine-Gaussian (also called Morlet-Gabor) wavelets to reconstruct the non-Gaussian features in the detector data, regardless of their origins. Even though sine-Gaussians wavelets form a non-orthogonal frames¹, their shape is highly variable and can optimally reconstruct a gravitational-wave signal with no *a priori* assumption on the signal source or morphology. Figure 2.1 shows how a waveform is reconstructed by summing over multiple sine-Gaussian wavelets.

In *BayesWave*, each wavelet in the time domain has five intrinsic parameters t_0, f_0, Q, A, ϕ_0 which represents central time, central frequency, quality factor, amplitude and phase offset respectively. These intrinsic parameters can be expressed as a single parameter vector such that $\lambda = \{t_0, f_0, Q, A, \phi_0\}$ and the mathematical representation of a sine-Gaussian wavelet is given by

$$\Psi(t; t_0, f_0, Q, A, \phi_0) = Ae^{-\Delta t^2/\tau^2} \cos(2\pi f_0 \Delta t + \phi_0) \quad (2.1)$$

where $\tau \equiv Q/(2\pi f_0)$ and $\Delta t \equiv t - t_0$ [45].

¹Discrete wavelets can form orthogonal bases for signal or glitch representations, but projecting the signal wavelets onto each detector requires the time translation operator which is computationally expensive. Despite the lack of orthogonality, sine-Gaussian wavelets are flexible in shape and have an analytic Fourier representation. Hence the analysis can be done in the frequency domain without the need of a time-translation operator. Further details can be found in Section 3 of [22].

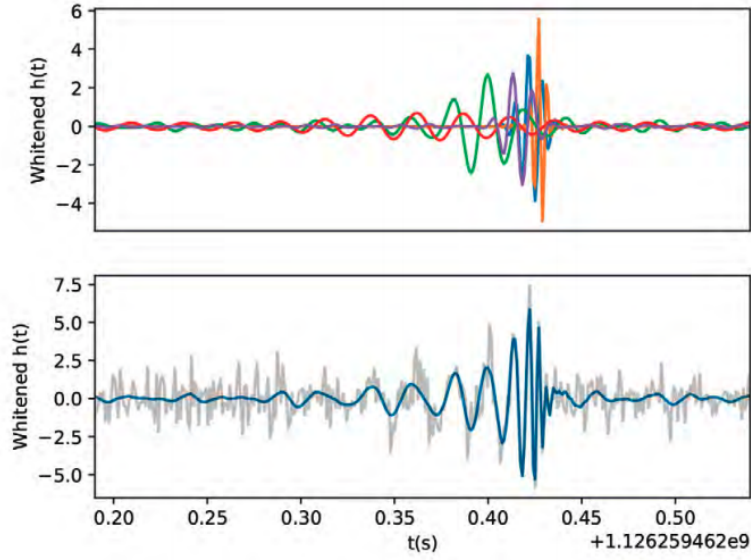


Figure 2.1: The top panel shows individual sine-Gaussian wavelets, each of different colour, used in the reconstruction of the first detection event GW150914 in the time-domain. The bottom panel shows the resulting waveform as a sum of all wavelets, overlaid on the actual data shown in grey. [Image adapted from: [LIGO Magazine, Issue 16, March 2020](#)]

Glitch reconstructions are unique for each detector owing to the fact that noise artefacts are non-identical and independent across different detectors. Hence, the set of glitch model parameters across a detector network must contain the respective parameters of glitch reconstruction for each individual detector. For instance, the complete set of glitch model parameters for the HLV network can be written as [45]

$$\theta_{\mathcal{G}} = \{\lambda^H \cup \lambda^L \cup \lambda^V\} \quad (2.2)$$

with $\lambda^i = \{\lambda_0^i \cup \lambda_1^i \cup \dots \cup \lambda_{N_{\mathcal{G}}^i}^i\}$ where the numerical subscripts indicate a single wavelet used in the glitch model and $N_{\mathcal{G}}^i$ is the total number of wavelets in the glitch model. The subscripts and superscripts indicates the i -th detector in the network.

In contrast to the glitch model, the signal model is common across all detectors in the network. As a result, a signal model should have a geocentric set of intrinsic wavelet parameter $\lambda^{\oplus} = \{\lambda_0 \cup \lambda_1 \cup \dots \cup \lambda_{N^{\mathcal{S}}}\}$ applicable throughout the network, where $N^{\mathcal{S}}$ is the number of wavelets used to fit the signal model. Additionally, signal models require a set of extrinsic parameter which governs the projection of the signal onto the detector network, or in other words, they provide information on what the true waveform, $h(t)$ looks like in each detector [45]. The extrinsic parameter vector is given by $\Omega = \{\alpha, \delta, \psi, \epsilon\}$ which sequentially contains the right ascension (RA), declination (dec), polarisation angle and ellipticity of the gravitational wave signal. The sky location (RA, dec) and polarisation angle of a source determine antenna beam patterns of the detector network, as well as provide information on the amplitude and the arrival-time delay of the signal in each detector [18]. Ellipticity defines the relative phase and amplitude of the plus and cross polarisations, h_+ and $h_{\times}$ respectively. In the frequency domain, the two polarisations are related by $h_{\times} = \epsilon h_+ e^{i\pi/2}$ where the ellipticity parameter ϵ takes values between 0 to 1 with the lower and upper bounds denoting linear to circular polarisations respectively [22]. Altogether a complete set of signal model parameters is

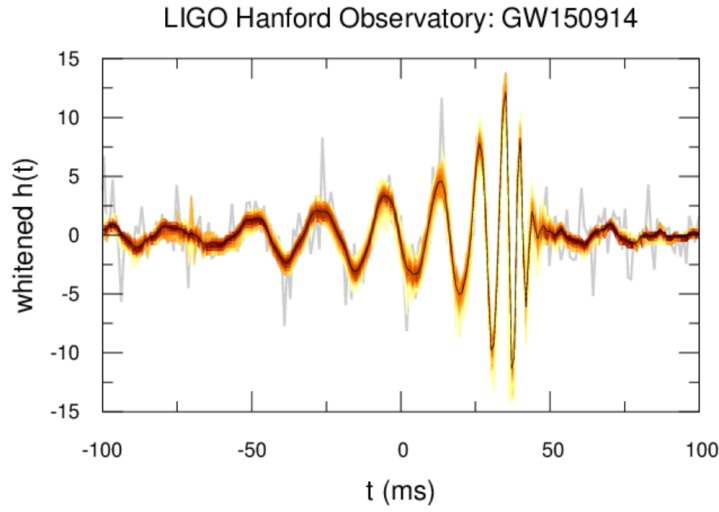


Figure 2.2: Time-domain reconstruction of GW150914. The coloured region is a two-dimensional histogram (posterior distribution) produced by combining all the *BayesWave*-reconstructed signal waveforms. Darker regions indicate higher likelihood of the reconstructed waveform matching the data. [Image courtesy of Jonah Kanner, Tyson Littenberg, and Meg Millhouse]

given by [45]

$$\theta_{\mathcal{S}} = \{\lambda^{\oplus} \cup \Omega\}. \quad (2.3)$$

The number of wavelets, N used in the reconstruction is marginalised over via a trans-dimensional Markov Chain Monte Carlo algorithm, otherwise known as the Reversible-Jump Markov chain Monte Carlo (RJMCMC), where signals with complex structure in time-frequency plane use more wavelets in the reconstruction. Previous studies [45, 50] have shown that N scales linearly with SNR such that

$$N \approx \eta + \gamma \text{ SNR} \quad (2.4)$$

where η and γ are constants which depend on waveform morphology. It was found that η , γ and hence N increases with waveform complexity [45]. For BBH systems with equal component masses of $50M_{\odot}$, the typical numbers were found to be $\eta = 5.6$ and $\gamma = 0.066$ for sine-Gaussian wavelet reconstructions [50].

2.2 Waveform Posteriors

BayesWave produces posterior distributions of the wavelet parameters described above. Each draw from the posterior contains a unique set of wavelet parameters (and extrinsic parameters for the signal model), which are then summed to produce a posterior of the waveform, $h(t)$. To visualise this, Figure 2.2 shows the posterior distribution of the time-domain reconstruction of the BBH waveform from event GW150914. By using sine-Gaussian wavelet frames, non-Gaussian waveform in the data is reconstructed with no *a priori* assumption on its source. We now discuss how *BayesWave* obtains the posterior distributions.

Given that each model $\mathcal{M}_j$ is associated with a set of parameters, $\vec{\theta}_j$, and $\vec{s}$ is the set of observed data which the model should fit, the joint posterior distribution of the model parameters $\vec{\theta}_j$ can be obtained through Bayes' Theorem

$$p(\vec{\theta}_j|\vec{s}, \mathcal{M}_j) = \frac{p(\vec{\theta}_j|\mathcal{M}_j)p(\vec{s}|\vec{\theta}_j, \mathcal{M}_j)}{p(\vec{s}|\mathcal{M}_j)}. \quad (2.5)$$

Here, $p(\vec{\theta}_j|\mathcal{M}_j)$ is the prior probability of parameters $\vec{\theta}_j$ before the data was observed, $p(\vec{s}|\vec{\theta}_j, \mathcal{M}_j)$ is the likelihood of obtaining the observed data given the model $\mathcal{M}_j$ and parameters $\vec{\theta}_j$. $p(\vec{s}|\mathcal{M}_j)$ is the model evidence (also called the marginalised evidence) given by [55]

$$p(\vec{s}|\mathcal{M}_j) = \int p(\vec{\theta}_j|\mathcal{M}_j)p(\vec{s}|\vec{\theta}_j, \mathcal{M}_j)d\vec{\theta}_j. \quad (2.6)$$

One can gather from Equation 2.6 that the model evidence will be a multivariable integral for models with various parameters and such integrals requires extreme computational power for direct evaluation. To avoid such complexity, *BayesWave* uses an alternative numerical approach - the RJMCMC to estimate the posterior distribution of signal and glitch models that fit the observed data, without having to directly compute the evidence. The algorithm thoroughly explores different model spaces and their respective parameter spaces by allowing for transition between model dimensions (number of wavelets). Just like the standard MCMC algorithm, the estimated posterior probability of a given model corresponds to the frequency of chain visits to the given model in the model space [22, 33].

In order to optimise the convergence time of the MCMC algorithm, the chosen proposal distribution from which the algorithm samples should match the posterior distribution closely. However, it is difficult to guess what the proposal distribution should be without knowing how the posterior looks like. For that reason, the parallel tempering approach is introduced in the *BayesWave* algorithm to allow for multiple chains to run simultaneously at different "temperatures", T . This parameter modifies the likelihood distribution of the target distribution such that [44]

$$p(\vec{\theta}_j|\mathcal{M}_j) = p(\vec{\theta}_j)p(s|\theta_j)^\beta \quad (2.7)$$

where $\beta = 1/T$. Chains at high temperatures (low β) are hence able to more easily explore the entire prior volume as opposed to cold chains that can only map the peak of the posterior distribution. Information is communicated between the hot and cold chains which then allows the whole prior volume to be thoroughly explored in a more efficient way. Even though only points in the $\beta = 1$ chain sample the true posterior, parallel tempering encourages quicker convergence to the target distribution [22].

2.3 Bayes Factor

The detection statistic associated with *BayesWave* is the Bayes factor, which compares two competing hypothesis via the strengths of their evidences. In *BayesWave*, the hypotheses are the different types of waveform models (signal, glitch or noise). The Bayes factor is discussed in further detail in Section

2.3.1. In this work, we aim to use the Bayes factor between signal and glitch models to quantify detection confidence between detector networks of different sizes. Thus it is in our best interest to mathematically understand the conditions of Bayesian model selection. We want to know under what circumstances a model is favoured over another, in other words, what factors influence changes to the Bayes factor. For that reason, we discuss the mathematical implications of Occam's Razor on the Bayes factor in Section 2.3.2 and analytically approximate a scaling relation for the Bayes factor in Section 2.3.3.

2.3.1 Bayes factor for Model Selection

The ratio of model evidences is the key to model selection in Bayesian inference as it assesses the plausibility of two different models, $\mathcal{M}_0$ and $\mathcal{M}_1$, parametrised by their respective parameter sets $\vec{\theta}_0$ and $\vec{\theta}_1$. This ratio is otherwise known as the Bayes factor,

$$\mathcal{B}_{1,0}(\vec{s}) = \frac{\int p(\vec{\theta}_1|\mathcal{M}_1)p(\vec{s}|\vec{\theta}_1, \mathcal{M}_1)d\vec{\theta}_1}{\int p(\vec{\theta}_0|\mathcal{M}_0)p(\vec{s}|\vec{\theta}_0, \mathcal{M}_0)d\vec{\theta}_0} \quad (2.8)$$

$$= \frac{p(\vec{s}|\mathcal{M}_1)}{p(\vec{s}|\mathcal{M}_0)}. \quad (2.9)$$

which follows from the definition of model evidence in Equation 2.6. The Bayes factor can be thought of as the betting odds between the two models, assuming equal prior probabilities on each model. $\mathcal{B}_{1,0}(\vec{s}) > 1$ implies that model $\mathcal{M}_1$ is more strongly supported by the data in comparison to model $\mathcal{M}_0$. In other words, $\mathcal{M}_1$ is the more appropriate representation of the waveform present in the detector data.

As previously mentioned, computing the model evidence by directly evaluating the integral in Equation 2.6 is extremely difficult and computationally expensive. Providentially, the parameter β introduced in parallel tempering can also be used to calculate the evidence through thermodynamic integration. The first step is to consider the modified posterior distribution in Equation 2.7 as part of some partition function

$$Z(\beta) \equiv \int d\vec{\theta} p(\vec{s}|\vec{\theta}, \mathcal{M})^\beta p(\vec{\theta}). \quad (2.10)$$

Since the prior is independent of β the integral can be rewritten as

$$\frac{d}{d\beta} \ln Z(\beta) = \left\langle \ln p(\vec{s}|\vec{\theta}, \mathcal{M}) \right\rangle_\beta \quad (2.11)$$

where the right hand side is the expectation value of the likelihood for the chain with temperature $T = 1/\beta$. Integrating the expectation value over β gives the log evidence

$$\ln p(\vec{s}|\mathcal{M}) = \int_0^1 d\beta \left\langle \ln p(\vec{s}|\vec{\theta}, \mathcal{M}) \right\rangle_\beta \quad (2.12)$$

which can then be used to calculate the log Bayes factor. By changing the integral variables from $\beta = \ln \beta$ the log Bayes factor between the two models can be written as [44, 56]

$$\ln \mathcal{B}_{1,0} = \int_{-\infty}^0 \beta \left(\left\langle \ln p(\vec{s}|\vec{\theta}_1, \mathcal{M}_1) \right\rangle_\beta - \left\langle \ln p(\vec{s}|\vec{\theta}_0, \mathcal{M}_0) \right\rangle_\beta \right) d \ln \beta. \quad (2.13)$$

2.3.2 Occam Penalty

A key to understanding Bayes factor behaviour when using a trans-dimensional model as *BayesWave* does, is to understand the role of the Occam penalty. We do this by looking at the overall behaviour of model evidences which directly influences the Bayes factor.

The parameter value at which the posterior distribution peaks is known as the maximum *a posteriori* (MAP), and is denoted as $\vec{\theta}_{\text{MAP}}$. For high signal-to-noise ratio (SNR) events, the integrand of model evidence in Equation 2.6 peaks sharply in the vicinity of the MAP. According to reference [45], an event should have $\text{SNR} \gtrsim 10$ to be considered a strong detection candidate and hence suitable for estimating its model evidences using the Laplace-Fisher approximation. Following the approximation, the integral in Equation 2.6 can be estimated as

$$p(\vec{s}|\mathcal{M}) \simeq p(\vec{s}|\vec{\theta}_{\text{MAP}}, \mathcal{M})p(\vec{\theta}_{\text{MAP}}|\mathcal{M})(2\pi)^{D/2}\sqrt{\det C}. \quad (2.14)$$

where $p(\vec{s}|\vec{\theta}_{\text{MAP}}, \mathcal{M})$ is the likelihood evaluated at the MAP parameters; $p(\vec{\theta}_{\text{MAP}}|\mathcal{M})$ is the prior evaluated at the MAP parameter values; D model dimension; and $\det C$ is the determinant for the full covariance matrix of reconstructed waveform parameters. If the covariance matrix for the parameters of a single wavelet is C_n , then assuming minimal overlap between parameter spaces of each wavelet:

$$\det C = \prod_{n=1}^N \det C_n. \quad (2.15)$$

Since the Laplace-Fisher approximation is associated with the MAP likelihood, C_n can be approximated by taking the inverse of the corresponding Fisher Information Matrix² (FIM) of the wavelet, Γ . The FIM of each wavelet is written in terms of its intrinsic parameters $\{t_0, f_0, Q, A, \phi_0\}$ [9, 22]

$$\Gamma = \text{SNR}^2 \begin{pmatrix} \frac{4\pi^2 f_0^2 (1+Q^2)}{Q^2} & 0 & 0 & 0 & -2\pi f_0 \\ 0 & \frac{3+Q^2}{4f_0^2} & -\frac{3}{4Qf_0} & -\frac{1}{2f_0} & 0 \\ 0 & -\frac{3}{4Qf_0} & \frac{3}{4Q^2} & \frac{1}{2Q} & 0 \\ 0 & -\frac{1}{2f_0} & \frac{1}{2Q} & 1 & 0 \\ -2\pi f_0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.16)$$

By definition, the determinant of a covariance matrix, $\det C$ measures the variance of the likelihood. Thus, $\sqrt{\det C}$ quantifies the characteristic spread of the likelihood function. The product of $\sqrt{\det C}$ and $(2\pi)^{D/2}$, which account for the dimensionality of the model, can then be used as a measure for the volume of the uncertainty ellipsoid (posterior volume), $\Delta V_{\mathcal{M}}$ for a given model $\mathcal{M}$ [43, 45, 55]. Assuming uniform priors for all wavelet parameters, one can also write $p(\vec{\theta}_{\text{MAP}}|\mathcal{M}) = 1/V_{\mathcal{M}}$ where $V_{\mathcal{M}}$ represents the total parameter space volume. Hence, the last three factors of Equation 2.14 can collectively be interpreted as the fraction of the prior occupied by the posterior distribution, such that the model evidence is now given by

$$p(\vec{s}|\mathcal{M}) \simeq p(\vec{s}|\vec{\theta}_{\text{MAP}}, \mathcal{M}) \frac{\Delta V_{\mathcal{M}}}{V_{\mathcal{M}}}. \quad (2.17)$$

²*BayesWave* uses FIMs to propose and update wavelet parameters by drawing from a multivariate Gaussian distribution $q(\mathbf{y}|\mathbf{x}) = \frac{\det \Gamma}{(2\pi)^2} \exp\left(-\frac{1}{2}\Gamma_{ij}\Delta x^i \Delta x^j\right)$ where $\Delta x^i = x^i - y^i$ denote the displacements in intrinsic parameter i before and after the update. Since FIMs contain information on local curvature of the likelihood of wavelet parameters, it accelerates convergence to the target (posterior) distribution by proposing MCMC jumps towards regions of higher likelihood [22].

Following equations 2.9 and 2.17, the Bayes factor between two models can be re-expressed as

$$\mathcal{B}_{1,0}(\vec{s}) = \Lambda_{1,0}(\vec{s}) \frac{\Delta V_1}{V_1} \frac{V_0}{\Delta V_0} \quad (2.18)$$

where $\Lambda_{1,0}(\vec{s})$ denotes the ratio of MAP likelihoods

$$\Lambda_{1,0}(\vec{s}) = \frac{p(\vec{s}|\vec{\theta}_{\text{MAP},1})}{p(\vec{s}|\vec{\theta}_{\text{MAP},0})}. \quad (2.19)$$

Equation 2.18 suggests that the Bayes factor is nothing more than product of the MAP likelihood ratio between the two models and the ratio of their Occam penalty factors, $\Delta V_{\mathcal{M}}/V_{\mathcal{M}}$. In words, the Occam factor is the fraction of prior taken up by the posterior [45]. It penalises models that require an unnecessarily large parameter space volume to fit the data by suppressing the model evidence, resulting in the Bayes factor disfavouring the model. Say $\mathcal{M}_1$ and $\mathcal{M}_0$ equally represent a given dataset i.e. they have similar MAP likelihoods, $\Lambda_{1,0}(\vec{s}) \approx 1$ and similar posterior volumes, $\Delta V_1 \approx \Delta V_0$. Then if $\mathcal{M}_1$ requires a larger intrinsic parameter space volume to fit the data such that $V_1 > V_0$, then the Occam factor takes a value less than unity. This reduces the value of the Bayes factor such that the model selection favours $\mathcal{M}_0$, while penalising the more complex model, $\mathcal{M}_1$. This is related to Occam's razor, a philosophical problem-solving principle which argues that the simplest solution should take precedence. Note that the Occam penalty is not an intentionally added component to the Bayes factor, rather it is inherently a result of using the Bayes Theorem.

2.3.3 Bayes Factor Scaling

The previous section showed that for two models that fit the data equally as well, the model that has a smaller dimensionality will be preferred. To get an intuitive understanding of this in the context of *BayesWave*, recall that signal models ($\mathcal{S}$) for each detector share the same set of intrinsic parameters along with four extrinsic parameters. Since there are five intrinsic parameters, (t_0, f_0, Q, A, ϕ_0) per wavelet, the dimension of signal models scales as $D_{\mathcal{S}} \sim 5N + 4$ where N is the number of wavelets. The glitch model ($\mathcal{G}$), on the other hand, have no extrinsic parameters but the glitch model of each detector is described by a unique set of intrinsic parameters. By assuming that glitch models in each detector will use similar number of wavelets as the signal models i.e. $N^{\mathcal{G}} \approx N^{\mathcal{S}} \equiv N$ (see Equation 2.26), the dimension of glitch model for a network of $\mathcal{I}$ detectors then scales as $D_{\mathcal{G}} \sim 5N\mathcal{I}$ [22]. From the scaling relations of $D_{\mathcal{S}}$ and $D_{\mathcal{G}}$, one can easily see that increasing the number of detectors will result in glitch models having more dimensions and hence higher complexity in comparison to signal models. If the signal and glitch models fit the data equally well, then by Occam's razor we should expect to see a selection bias towards the signal models with additional detectors in the network.

In Reference [45], Littenberg et al. put forth an analytic scaling of the log signal-to-glitch Bayes factor, $\ln \mathcal{B}_{\mathcal{S}\mathcal{G}}$, in an effort to fully understand *BayesWave's* ability to robustly distinguish astrophysical signals from instrumental glitches. This work showed that the primary scaling of the Bayes factor in a two-detector network goes as $\ln \mathcal{B}_{\mathcal{S}\mathcal{G}} \sim N \ln(\text{SNR}_{\text{net}})$, where N is the number of wavelets used in the reconstruction and SNR_{net} is the network injected SNR. Figure 2.3 displays the results from Reference [45] where injections with $\text{SNR}_{\text{net}} \geq 10$ show good agreement between the analytic log

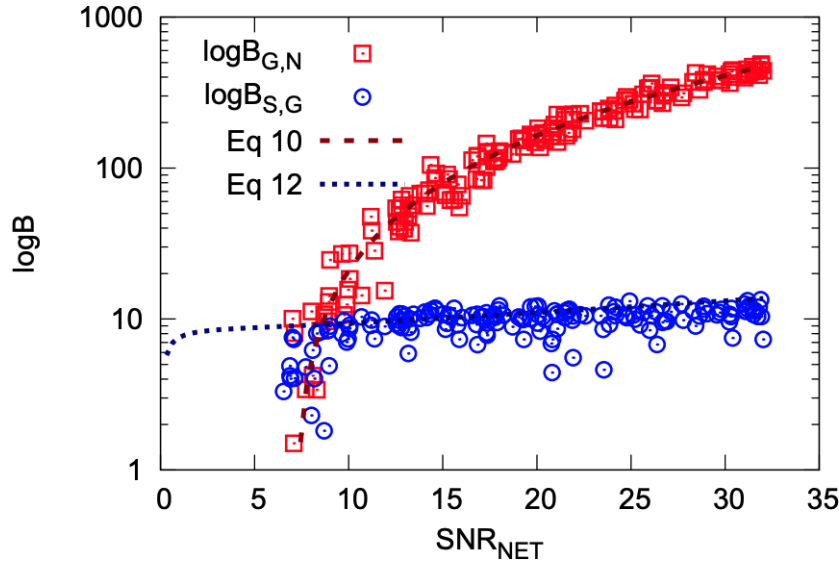


Figure 2.3: Comparing numerical results for log signal-to-glitch Bayes factor $\ln \mathcal{B}_{S,G}$ (blue circles) and log glitch-to-noise Bayes factor $\ln \mathcal{B}_{G,N}$ (red squares) to the relevant analytic predictions as indicated by the respective dotted lines. We only discuss the results for $\ln \mathcal{B}_{S,G}$ which is relevant to this study. Equation 12 is the two-detector ($\mathcal{I} = 2$) log Bayes factor equation derived from Laplace approximations of model evidences, and the blue circles indicate empirical $\ln \mathcal{B}_{S,G}$ obtained from recovering injected simulated sine-Gaussian gravitational wave signals from Gaussian noise with *BayesWave*. This result is taken directly from Figure 1 of Reference [45].

Bayes factor scaling with SNR for a two-detector network, derived using Laplace approximation to the evidence and the empirical Bayes factor recovered from the HL network using *BayesWave*. The injected waveforms for this analysis were sine-Gaussians, which are typically recovered by *BayesWave* with $N = 1$ wavelet. The dependence of Bayes factor on number of wavelets (and therefore the complexity of the signal in time frequency plane) differentiates *BayesWave* from other unmodelled searches whose detection statistics scale primarily with SNR. Here we extend this work to find the Bayes factor scaling with the number of detectors in the network (i.e. for $\mathcal{I} > 2$).

From the Laplace approximation for model evidences in Equation 2.14, one can derive the log evidences for the signal and glitch models

$$\ln p(\vec{s} | \mathcal{S}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \frac{5N^{\mathcal{S}}}{2} - N^{\mathcal{S}} \ln(V_{\lambda}) + \sum_{n=1}^{N^{\mathcal{S}}} \ln \left(\frac{\bar{Q}_n}{\text{SNR}_{\text{net},n}^5} \right) + \frac{D_{\Omega}}{2} + \ln \frac{\sqrt{\det C_{\Omega}}}{V_{\Omega}} \quad (2.20)$$

$$\ln p(\vec{s} | \mathcal{G}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \sum_{i=1}^{\mathcal{I}} \left[\frac{5N_i^{\mathcal{G}}}{2} + N_i^{\mathcal{G}} \ln(V_{\lambda}) - \sum_{n=1}^{N_i^{\mathcal{G}}} \ln \left(\frac{\bar{Q}_n}{\text{SNR}_{i,n}^5} \right) \right] \quad (2.21)$$

where $\bar{Q}_n = (2\pi)^{5/2} \frac{\sqrt{2}Q_n}{\pi}$ and N^x is the total number of wavelets for model x (i.e. signal or glitch). Subscript i refers to detector i in the network and n labels an individual wavelet from the set of wavelets for a given model, for instance $\text{SNR}_{i,n}$ is the SNR of wavelet n in the i -th detector³. In the last two terms of Equation 2.20, D_{Ω} , C_{Ω} and V_{Ω} denote the dimension, covariance matrix and the

³Each individual wavelet used in signal or glitch model reconstruction has an amplitude which can be converted into to SNR as described in Reference [22] (see Equation 3.3).

prior volume of extrinsic parameters respectively. Since there are four extrinsic parameters, $D_\Omega = 4$. Full derivations to Equations 2.20 and 2.21 can be found in Appendix A [45].

The difference between the two log model evidences gives the log Bayes factor between the signal and glitch models

$$\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} = \ln p(\vec{s}|\mathcal{S}) - \ln p(\vec{s}|\mathcal{G}). \quad (2.22)$$

However, to see how the Bayes factor scales with the waveform parameters, some assumptions can be made to simplify the two equations above and the assumptions are taken directly from [45]. Loud signals typically have optimal extrinsic parameters across the detector network so each detector receives similar signal strength such that

$$\text{SNR}_{i,n} \approx \frac{\text{SNR}_{\text{net},n}}{\sqrt{\mathcal{I}}}. \quad (2.23)$$

We also assume that the SNR of each wavelet is the same such that

$$\text{SNR}_{\text{net},n} \approx \frac{\text{SNR}_{\text{net}}}{\sqrt{N}}. \quad (2.24)$$

In this case, the glitch model will use similar reconstruction parameters as the signal model such that the quality factor of all wavelets are approximately equal

$$Q_{i,n}^{\mathcal{G}} \approx Q_n^{\mathcal{S}} \equiv Q. \quad (2.25)$$

Similarly for the number of wavelets

$$N^{\mathcal{G}} \approx N^{\mathcal{S}} \equiv N. \quad (2.26)$$

Thus Equations 2.20 and 2.21 can be simplified to

$$\ln p(\vec{s}|\mathcal{S}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \frac{5N}{2} - N \ln(V_\lambda) + \sum_{n=1}^N \ln(\bar{Q}_n) - 5N \ln\left(\frac{\text{SNR}_{\text{net}}}{\sqrt{N}}\right) + \left(2 + \ln \frac{\sqrt{\det C_\Omega}}{V_\Omega}\right) \quad (2.27)$$

$$\ln p(\vec{s}|\mathcal{G}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \mathcal{I} \left[\frac{5N}{2} + N \ln(V_\lambda) - \sum_{n=1}^N \ln(\bar{Q}_n) + 5N \ln\left(\frac{\text{SNR}_{\text{net}}}{\sqrt{\mathcal{I}N}}\right) \right]. \quad (2.28)$$

and the theoretical log Bayes factor between the signal and glitch model for a network of $\mathcal{I}$ detector(s) can be estimated as follows

$$\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} \simeq (\mathcal{I} - 1) \left[\frac{5N}{2} + N \ln(V_\lambda) - \sum_{n=1}^N \ln(\bar{Q}_n) + 5N \ln\left(\frac{\text{SNR}_{\text{net}}}{\sqrt{N}}\right) \right] - \frac{5}{2} \mathcal{I} N \ln(\mathcal{I}) + \left(2 + \ln \frac{\sqrt{\det C_\Omega}}{V_\Omega}\right). \quad (2.29)$$

The equation shows explicit dependence of the Bayes factor on the network SNR, number of wavelets and the number of detectors in the network. We pay close attention to the scaling

$$\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} \sim \mathcal{O}[\mathcal{I} N \ln \text{SNR}_{\text{net}}] \quad (2.30)$$

which implies that signal models at fixed SNR_{net} is favoured over the glitch model with increasing

complexity (higher N) and number of detectors (bigger $\mathcal{I}$). In other words, waveforms that are coincident across multiple detectors with a more complex structure in time-frequency plane are more likely to have physical significance, and hence have a higher likelihood of being classified as signal [45]. The scaling relation $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} \sim \mathcal{O}[\mathcal{I}]$ suggests that signal models are favoured over glitch models with increasing number of detectors for waveforms of equal SNR_{net} and complexity N . This analytic result agrees with the discussion at the start of this section where if two models fit the data equally well, the less complex model is more likely to be selected to represent the waveform.

However, if *BayesWave* is unable to construct signal models which fit the data of all detectors across the network, the MAP likelihood of signal models will be small compared to that of the glitch models. According to Equation 2.18, the Bayes factor then favours glitch model hence indicating low detection confidence for a physical signal.

2.4 Network Overlap

Recall that *BayesWave* produces posterior distributions of reconstructed waveforms (i.e. signal and glitch models) that represents the true waveform, $h(t)$. One way to quantify the agreement between the true signal waveform and the ones recovered by *BayesWave* is through the overlap

$$\mathcal{O} = \frac{(h_{\text{rec}} | h_{\text{inj}})}{\sqrt{(h_{\text{rec}} | h_{\text{rec}}) (h_{\text{inj}} | h_{\text{inj}})}}. \quad (2.31)$$

To emphasize that we are recovering injected waveforms with *BayesWave* in this work, we label the true waveform as h_{inj} , and h_{rec} denotes the waveform recovered by *BayesWave*.

However, Equation 2.31 only applies to a single detector. An overall network overlap, $\mathcal{O}_{\text{net}}$ is required to fully evaluate *BayesWave*'s performance in recovering waveforms from all the detectors combined. In order to define such quantity, we first note that even for a true signal, the waveforms will not be identical across the network because the detectors are at geographically separated and have different orientations. In other words, h_{inj} can be shifted temporally and/or appear differently in each detector depending on the direction of propagation (sky location) of the signal. For that reason, h_{inj} and h_{rec} are unique to each detector such that inter-detector waveform overlaps do not exist. Therefore we can simply take the sum of noise-weighted inner products over all detectors with the appropriate normalisation to obtain $\mathcal{O}_{\text{net}}$. For a network with $\mathcal{I}$ -detectors, we write

$$\mathcal{O}_{\text{net}} = \frac{\sum_{i=1}^{\mathcal{I}} (h_{\text{rec}}^{(i)} | h_{\text{inj}}^{(i)})}{\sqrt{\sum_{i=1}^{\mathcal{I}} (h_{\text{rec}}^{(i)} | h_{\text{rec}}^{(i)}) \sum_{i=1}^{\mathcal{I}} (h_{\text{inj}}^{(i)} | h_{\text{inj}}^{(i)})}} \quad (2.32)$$

where $h^{(i)}$ indicates waveforms for detector i .

Chapter 3

Waveform Injections

This research aims to compare the performance of *BayesWave* in recovering injected waveforms from noise for detector networks of different sizes. In this chapter, we discuss the methods of injecting different families of waveform into simulated detector noise. Section 3.1 describes the properties of the simulated data for the detectors relevant to this study. The following sections discuss families of injected waveforms and the methods of injection. The three waveforms used in this study are sine-Gaussian signals (Section 3.2), sine-Gaussian glitches (Section 3.3) and phenomenological binary black hole (BBH) waveforms (Section 3.4).

3.1 Simulated Detector Data

In this work, we inject waveforms of three different families into simulated Gaussian noise at projected LIGO, Virgo and KAGRA sensitivities for the fourth observing run, O4, as given in the LIGO, Virgo and KAGRA Observing Scenarios [7]. We show the sensitivity curves in Figure 3.1, where strain noise in the vertical axis is the root of the one-sided PSD, $\sqrt{S_n(f)}$ or otherwise known as amplitude spectral density (ASD). The ASD has units of inverse frequency, Hz^{-1} which came from the definition of dimensionless noise amplitude¹, H_n given by [51]

$$[H_n(f)]^2 = f S_n(f). \quad (3.1)$$

We can also compare the sensitivities of the detectors with a figure of merit known as the range, R of characteristic CBCs. This quantity corresponds to the average distance at which a CBC merger can be detected and is typically given in units of megaparsecs (Mpc). In the literature from which we obtained our noise curves shown in Figure 3.1, the binary neutron star (BNS) range was used to quantify detector performance² can be computed by rearranging the expression

$$V = \frac{4\pi}{3} R^3 \quad (3.2)$$

where V is the comoving volume of the orientation-averaged spacetime volume surveyed by a detector per unit detector time where each detector is assumed to have an matched-filter detection SNR threshold of 8. Comoving here implies that the volume is multiplied by a factor of $1 + z$ which corrects for time dilation. In O3, the LIGO Hanford (H), LIGO Livingston (L), Virgo (V) and KAGRA (K)

¹The counterpart of noise amplitude which describes signal amplitude at different frequencies is known as the characteristic strain, $[H_c(f)]^2 = 4f^2 |\tilde{h}(f)|^2$.

²The version of Observing Scenarios referenced in this work is <https://dcc.ligo.org/LIGO-P1200087/public>.

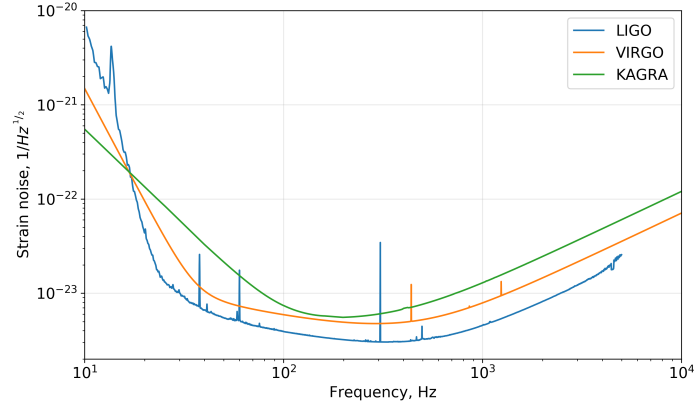


Figure 3.1: Simulated O4 LIGO, Virgo and KAGRA strain sensitivities, $\sqrt{S_n}$ as a function of frequency. (Data from: <https://dcc.ligo.org/LIGO-T2000012/public>).

detectors had BNS ranges of 110 Mpc, 130 Mpc, 50 Mpc and 8 – 25 Mpc respectively. In the same detector order for O4, the expected BNS ranges are 160 – 190 Mpc (both H and L), 90 – 120 Mpc and 25 – 130 Mpc [7]. Despite the large uncertainty in KAGRA’s O4 projected sensitivity, all three detectors are expected to show improvements in overall sensitivity compare to that in O3.

3.2 Sine-Gaussian Signal Injections

First we inject single sine-Gaussian wavelets as coincident signals into simulated noise of the HL, HLV and HLKV networks. The objective of injecting single sine-Gaussian wavelets is to test the consistency of *BayesWave* in waveform recovery and its ability to identify signal waveforms from the combined data of different detector networks. In addition, the number of wavelets used to reconstruct single sine-Gaussian wavelets do not change (i.e. always $N = 1$), so it is much easier to verify the analytic scaling of the Bayes factor with the number of detectors as per 2.30. We discuss the results and analysis in Section 4.1 of Chapter 4.

To construct a single sine-Gaussian wavelet (see Equation 2.1), we randomly draw four of its intrinsic parameters from uniform distributions of appropriate ranges where $t_0 \in [1.5, 2.5]$ s (a randomly selected range of GPS time), $f_0 \in [32, 1000]$ Hz, $Q \in [0.1, 40]$ and $\phi_0 \in [0, 2\pi]$. The amplitude of the wavelet, A , is obtained via the equation [50]

$$A = \text{SNR} \sqrt{\frac{2\sqrt{2\pi}f_0 S_n(f_0)}{Q}} \quad (3.3)$$

with the value SNR drawn randomly from a uniform distribution and $\text{SNR} \in [10, 50]$. Since we are performing a signal injection, we expect the wavelets to have four extrinsic parameter which are also drawn randomly from uniform distributions such that $\alpha \in [0, 2\pi]$, $\delta \in [-\pi/2, \pi/2]$, $\psi \in [0, 2\pi]$ and $\epsilon \in [-0.99, 0.99]$. For each complete set of wavelet parameters, *BayesWave* constructs and inject a sine-Gaussian wavelet into the noise data of a given detector network at different times (i.e. different trigger time for each injection), such that the injections are sufficiently far apart in time so that they do not overlap. An example of a wavelet in the time domain is shown in the left panel Figure 3.2 where the vertical axis shows the strain amplitude.

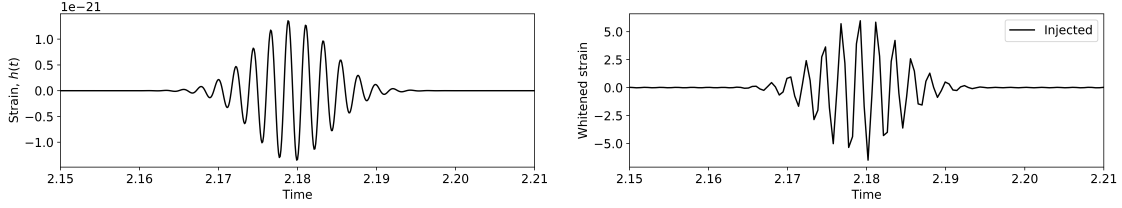


Figure 3.2: The left and right panels show the same sine-Gaussian wavelet in the time-domain before and after whitening. This wavelet is arbitrarily chosen and has intrinsic parameters: $t_0 = 2.18\text{s}$, $f_0 = 451.06\text{Hz}$, $A = 1.37 \times 10^{-21}$, $Q = 19.34$ and $\phi_0 = 1.19$.

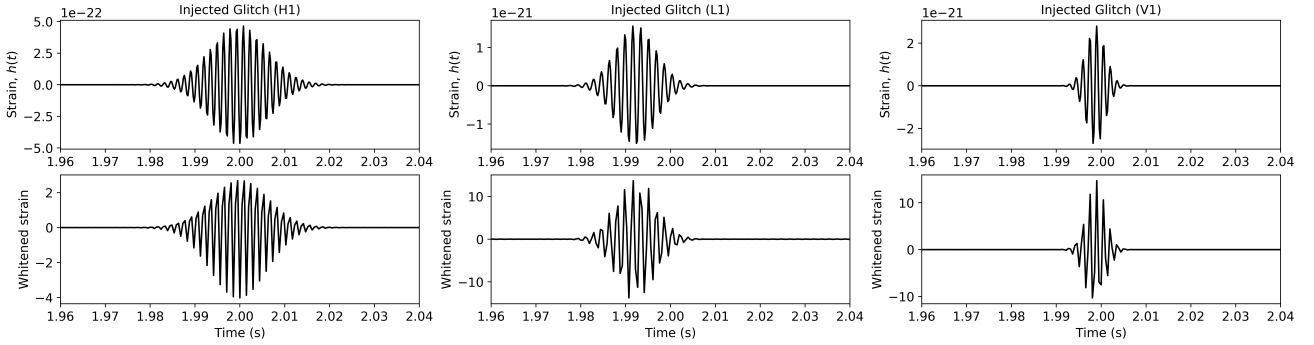


Figure 3.3: A set of non-identical, coincident glitches in the time domain across the HLV network. The top and bottom panels show the strain of the glitches before and after whitening.

In Figure 3.1, we see that the detector noise into which the wavelet is injected varies in loudness at different frequencies. Thus we show the whitened waveform to ensure that the data at each frequency bin are of equal significance despite the having louder noise at certain frequencies. Whitening requires *BayesWave* to Fourier transform (FFT) the data into the frequency domain and dividing by the ASD before inverse Fourier transforming (iFFT) it back into the time domain [64]

$$h(t) \xrightarrow{\text{FFT}} \tilde{h}(f) \xrightarrow{\text{whiten}} \tilde{h}_w(f) = \frac{\tilde{h}(f)}{\sqrt{S_n(f)}} \xrightarrow{\text{iFFT}} h_w(t) \quad (3.4)$$

The right panel of Figure 3.2 shows the same example wavelet after the whitening. Note that the whitened waveform plot appears less smooth due to binning of the time axis upon injection.

3.3 Sine-Gaussian Glitch Injections

We also want to test the reliability of *BayesWave* in identifying *glitches* when they happen to be coincident across all detectors in the network. We achieve this by injecting simultaneous, independent glitches into each detector and ensure that upon waveform recovery, *BayesWave* outputs a Bayes factor that favours glitch model over signal model regardless of the size of detector network. We also check for improvements in glitch identification (i.e. more negative Bayes factor) with increasing number of detectors in the network. We present the results in Section 4.2 of Chapter 4.

In this context, independent means that the parameters of each glitch are independently and randomly drawn for each detector from appropriate distributions. For simplicity, we use single sine-Gaussian wavelets to represent glitches. We draw the intrinsic wavelet parameters from the same distributions as the sine-Gaussian wavelets specified in Section 3.2, with the exception of the central

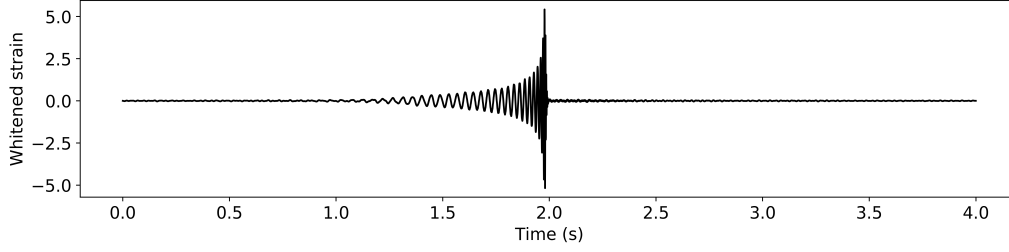


Figure 3.4: Sample simulated waveform of a BBH merger in the time domain. This waveform was whitened upon injection and was injected at $\text{SNR} = 28.78$ into the LIGO Hanford detector noise.

time, t_0 . For waveforms to be ‘coincident’, it must be present within a set time frame across all detectors. This time frame is determined by the distance between the detectors and the finite propagation speed of gravitational waves (speed of light). The detection delay between the two LIGO detectors is approximately 7 milliseconds (ms), but since our analysis involves the Virgo, we arbitrarily extend the coincident time frame to 20ms to account for the larger distances between the four detectors. Thus we choose to draw t_0 of each glitch from a uniform distribution within the range $t_0 \in [1.99, 2.01]$ s. We did not have to implement this for the sine-Gaussian injections because the same waveform (same t_0) is injected into all detectors. Figure 3.3 shows a set of simulated coincident glitches in the time domain across the HLV network. The glitches are plotted on the same time scale to emphasize that even though the central times are similar, their shapes and durations are highly variable across detectors, indicating that they are coincident but non-identical (i.e. constructed independently).

3.4 Binary Black Hole (BBH) Injections

In addition to sine-Gaussian waveforms, we also inject a set of simulated binary black hole (BBH) waveforms. BBHs are an abundant source of signals for LIGO and easy infrastructure exists to simulate these waveforms. Hence we can investigate how well *BayesWave* identifies them as signals (via the Bayes factor), and how well the signals are reconstructed (via the overlap and waveform moments) by recovering injected BBH signals from HL, HLV and HLKV networks using *BayesWave*. The results and analyses are presented in Sections Chapter 5.

We use tools from the LIGO Analysis Library (`lalapps_inspinj`) [42] to simulate non-spinning BBHs with equal component masses of $30M_\odot$. The masses and hence mass ratio of all BBH injections are kept constant to ensure that variations in the outcomes are only due to detector network sizes and SNR. The gravitational-wave sources are distributed isotropically across the sky, and uniformly in SNR within the range $\text{SNR} \in [10, 50]$. We use the phenomenological waveform `IMRPhenomD` to model the three phases of the non-precessing binary. Inspiral phases of the models were constructed using a combination of the effective-one-body formalism³ (EOB) and analytic post-Newtonian (PN) methods of up to 2.5PN order. The merger-ringdown phases were calibrated using Numerical Relativity (NR) simulations [36, 38]. Figure 3.4 shows a sample BBH waveform in the time domain.

³The effective one-body formalism for a two-body problem uses a non-perturbative, analytical approach to describe the motion and gravitational wave radiation of a BBH system of arbitrary mass ratio during its late inspiral phase. The general idea is to map a two-body system onto an effective one-body system by canonical transformation and study the motion of the system projected as a test particle in some effective external space-time metric. A more comprehensive description of the EOB formalism can be found in References [17] and [24].

Chapter 4

Sine-Gaussian Waveform Detection

In this chapter, we present the results of *BayesWave* waveform recovery for the sine-Gaussian signal and glitch injections described in Sections 3.2 and 3.3 respectively. The purpose of these injections are to check for consistency of the *BayesWave* algorithm in identifying signals and glitches from detector networks of different sizes. We discuss the methods of obtaining the data and present the analysis for the sine-Gaussian wavelet injections in Section 4.1 by showing a plot of $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}}$ versus SNR_{net} to compare the overall Bayes factor between the HL, HLV and HLKV networks. Similarly for coincident, sine-Gaussian glitch injections in Section 4.2, but only for the HL and HLV networks. In Section 4.3, we visually compare the signal and glitch models constructed by *BayesWave* with the injected waveforms, for both the sine-Gaussian wavelet and glitch injections.

4.1 Recovering Injected Sine-Gaussian Signals

In this section, we test for *BayesWave*'s ability to identify signal waveforms using the log signal-to-glitch Bayes factor, $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}}$ which is computed by taking the difference in log model evidences as in Equation 2.22. If *BayesWave* correctly identifies an injection as a signal upon waveform recovery, we see $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} > 0$ such that the signal model $\mathcal{S}$ is more strongly supported by the data than the glitch model $\mathcal{G}$. The dataset used this analysis is a set of 150 single sine-Gaussian wavelets injected as coincident signals into detector noise of the HL, HLV and HLKV network. Parameters of injected waveforms are randomly drawn from their respective uniform prior distributions as described in Section 3.2. Injections are made at different detector times with a sufficiently large separation interval so that the events do not coincide or overlap with one another temporally.

In Figure 4.1, we plot $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}}$ of each injection against SNR_{net} (i.e. injected SNR) for the HL, HLV and HLKV network injections. The plots show that *BayesWave* is indeed able to correctly identify the coherent sine-Gaussian wavelets as signals from all three detector networks, seeing as all but one injection with $\text{SNR} > 10$ satisfy $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}} > 0$. Since *BayesWave* is unable to differentiate low amplitude waveform features from detector noise, low SNR injections are easily overlooked and potentially identified as noise. This explains why we observe a few injections of $\text{SNR} < 10$ with negative $\ln \mathcal{B}_{\mathcal{S},\mathcal{G}}$. Since the injected signal here matches the basis function used by *BayesWave* (i.e. a sine-Gaussian), we expect that the typical number of wavelets used in the reconstruction will be 1, regardless of SNR. For each injection, we take the median number of wavelets, N used in signal waveform reconstruction of the posterior output. Averaging N over all 150 injections and rounding

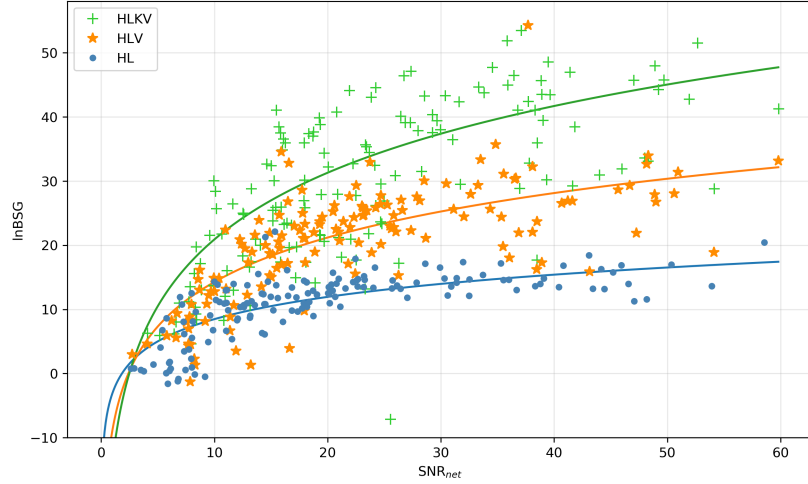


Figure 4.1: Log signal-to-glitch Bayes factor, $\ln \mathcal{B}_{S,G}$ of sine-Gaussian wavelet waveform recoveries versus network signal-to-noise ratio, SNR_{net} . The horizontal axis corresponds three different network SNRs: (i) for the blue dot symbols it corresponds to that of the HL network, (ii) for the orange star symbols it corresponds to that of the HLV network and (iii) for the green cross symbols it corresponds to that of the HLKV network. This same description applies to any axis labelled SNR_{net} in the figures to follow. The solid lines with colours corresponding to the data symbols are analytic predictions of $\ln \mathcal{B}_{S,G}$ given by Equation 4.1.

to the nearest integer, we find that for all three detector configurations, *BayesWave* uses $N = 1$ on average to reconstruct the injected sine-Gaussian wavelets. More precisely, *BayesWave* successfully identified 79% of the injections as sine-Gaussian wavelets by reconstructing them with $N = 1$. The remaining 21% are reconstructed with $N = 2$, which is still reasonable.

We also note that $\ln \mathcal{B}_{S,G}$ increases with SNR_{net} and is generally higher for larger detector networks. This observation agrees with the analytic scaling of $\ln \mathcal{B}_{S,G}$ shown in Equation 2.30. Since N is constant, we can be certain that the differences in $\ln \mathcal{B}_{S,G}$ between the different detector configurations at comparable SNR are entirely due to the number of detectors in network, $\mathcal{I}$ and so it allows us to analytically test the scaling of $\ln \mathcal{B}_{S,G}$ with $\mathcal{I}$. We attempt to fit the empirical results for each detector network with a generalised expression

$$\ln \mathcal{B}_{S,G} \approx (\mathcal{I} - 1)[5 \ln \text{SNR}_{\text{net}} + a] + \frac{5}{2} \mathcal{I} \ln \mathcal{I} + b. \quad (4.1)$$

which came directly from Equation 2.30 with $N = 1$. Since this is an approximation and that extrinsic parameters cannot be approximated analytically¹, we set $a = \frac{5N}{2} - N \ln(V_\lambda) + \sum_n^N \ln(\bar{Q}_n)$ and $b = 2 + \ln \frac{\sqrt{\det C_\Omega}}{V_\Omega}$ to be constants such that the prior volumes, V for both intrinsic and extrinsic parameters are equal for all detector configurations. The analytic fits are constructed by manually adjusting the values of a and b to fit the empirical data. It is not possible to perform an exact fit with the data since the scaling of $\ln \mathcal{B}_{S,G}$ with $\mathcal{I}$ assumes detectors of same sensitivities, whereas our data was obtained using detectors of varying sensitivities. We present the fits as three solid lines in Figure 4.1 with $a = -10$ and $b = 4$, where the colour of each line matches the colour of data points of the corresponding detector network. The same values of a and b are used for all three fits and they are consistent with the empirical results. This result is an extension to that of Reference [45] shown in Figure 2.3 such that we now consider networks with $\mathcal{I} > 2$.

¹FIM approximation for $\det C_\Omega$ is inadequate. The extrinsic parameter space contains degeneracy between parameters. This results in multimodal, non-Gaussian likelihood distribution which spans the full extent of the prior range [45].

To reiterate the takeaway of this analysis, *BayesWave* successfully reconstructed injected single sine-Gaussian wavelets with a single wavelet ($N = 1$) and identified them as signal waveforms with $\ln \mathcal{B}_{S,G} > 0$. Additionally, the agreement between empirical results and analytic predictions in this example suggests that the Bayes factor does not only depend on SNR_{net} but also $\mathcal{I}$. We make further claims on this speculation later in Section 5.1 with the result from BBH waveform recoveries.

4.2 Recovering Injected Glitches

We also evaluate *BayesWave*'s performance in identifying glitches by injecting coincident sets of glitches into the HL and HLV network. We excluded the HLKV network from this analysis since we are only testing if the *BayesWave* algorithm can correctly identify glitches. We inject a total of 50 detector-independent glitches into both networks. Glitch injections are different in each detector but are coincident across the detector network as described in Section 3.2. The same set of glitches were injected into the Hanford detector for both networks and similarly for the Livingston detector. However, the HLV network has Virgo as an additional detector, so we generate a third independent set of glitches just for the Virgo injections. The Virgo glitches are not present anywhere in the HL network.

We obtain $\ln \mathcal{B}_{S,G}$ of the glitch injection recoveries for both detector networks and plot the values against the respective SNR_{net} . The results are shown in the left panel of Figure 4.2, from which we can see that $\ln \mathcal{B}_{S,G} < 0$ for all injections in both detector networks. This implies that glitch models are more strongly supported by the data than signal models, or in other words, *BayesWave* has successfully identified all the injections as glitches rather than astrophysical signals.

SNR_{net} here is computed by taking the quadrature sum of injected glitch SNRs in each network just as we do for signal injections. However, SNR_{net} for glitches do not reflect the 'loudness' of a single waveform, rather it is the combined loudness of the uncorrelated glitches present across the detector network. Even though SNR_{net} in this context has no physical meaning, it is useful for gauging the overall performance of *BayesWave* in waveform recovery when glitches are present with varying loudness. Our results in the left panel of Figure 4.2 show that $\ln \mathcal{B}_{S,G}$ decreases (becomes more negative) with increasing SNR_{net} for both detector networks. This suggests that *BayesWave* can identify glitches with higher confidence when they are present across the network with higher SNR. Intuitively, discrepancies between dissimilar waveforms become more obvious at high SNR, thus it makes sense that *BayesWave* can more reliably regard them as independent glitches.

We also evaluate the performance of *BayesWave* in recovering glitches from detector networks of different sizes. We achieve this by comparing the $\ln \mathcal{B}_{S,G}$ between HL and HLV injections with the same glitches in Hanford and Livingston detectors. The comparison is made by plotting the HLV network $\ln \mathcal{B}_{S,G}$ versus that of the HL network and is shown in the right panel of Figure 4.2. The blue line indicates equal $\ln \mathcal{B}_{S,G}$ for the two networks. Since all data points lie below the line, we gather that $\ln \mathcal{B}_{S,G}$ is consistently lower (more negative) for the HLV network compared to the HL network. In other words, the HLV networks enables higher confidence glitch detections with *BayesWave* compared to the HL network. To explain this improvement, we first recall that $\ln \mathcal{B}_{S,G}$ is the difference between the log evidences of signal and glitch models. With Virgo containing a glitch that is coincident in time but dissimilar in shape to those in the Hanford and Livingston detectors,

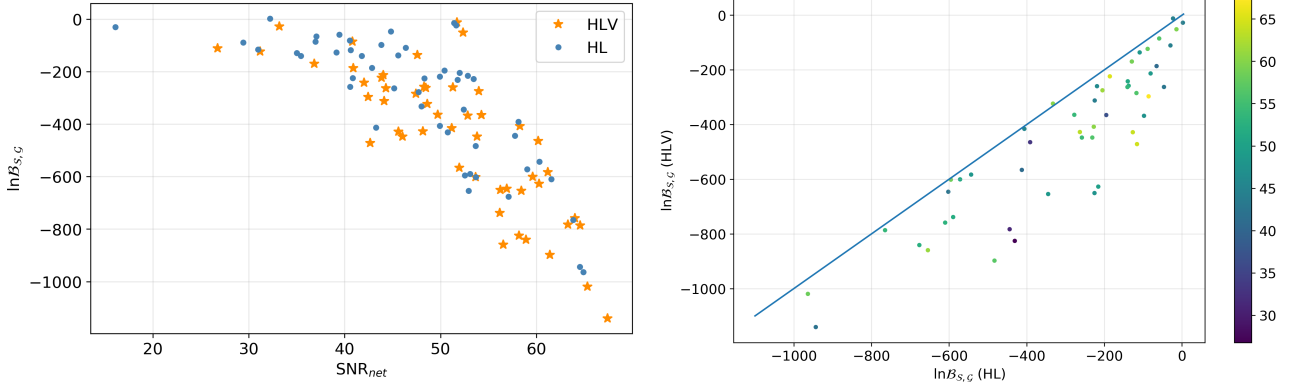


Figure 4.2: (Left panel) Log signal-to-glitch Bayes factor, $\ln \mathcal{B}_{S,G}$ of coincident glitch injection recoveries versus network signal-to-noise ratio, SNR_{net} . Horizontal axes is the same as that described in Figure 4.1. (Right panel) $\ln \mathcal{B}_{S,G}$ of the HLV network versus that of the HL network. The color bar shows the SNR_{net} of the HLV network for each injection.

the HLV signal model which tries to fit the data of all three detectors with the same waveform be a worse fit compared to the HL signal model which is only fitting data of two detectors. As the signal model is comparatively less consistent with the data than the corresponding glitch model, the log signal model evidence is also comparatively smaller than the log glitch model evidence for the HLV network which results in a more negative $\ln \mathcal{B}_{S,G}$ compared to the HL network.

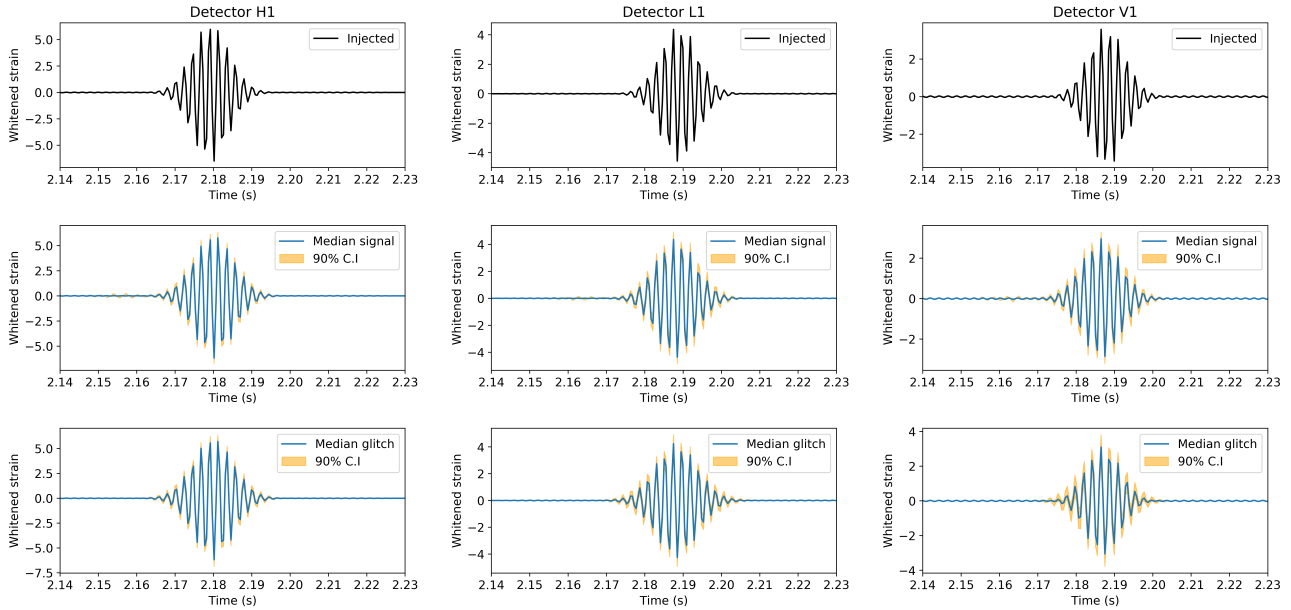
In summary, our results show that increasing SNR and expanding detector network enables higher confidence glitch identification with the *BayesWave* pipeline.

4.3 Comparing Signal and Glitch Models

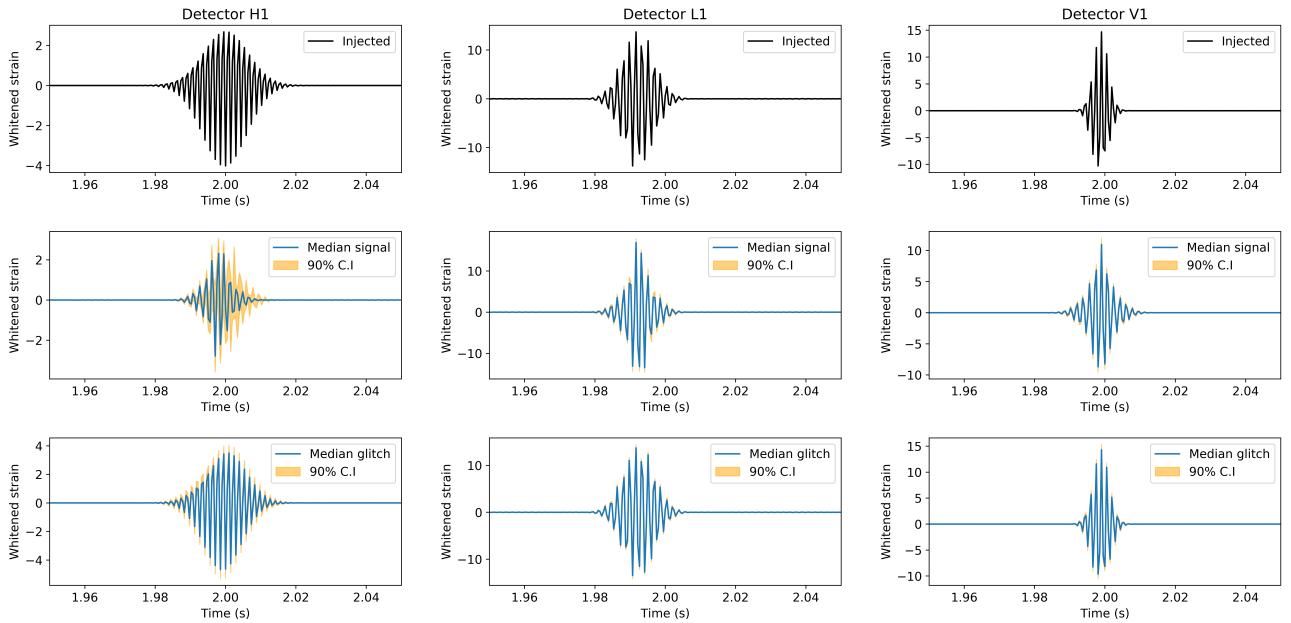
To visualise the results presented in Sections 4.1 and 4.2, we can compare the signal and glitch models constructed by *BayesWave* in the time domain with the injected waveform. In Figure 4.3a, we show the injected sine-Gaussian signals in each detector in the HLV network in the top row, along with the signal and glitch waveforms reconstructed by *BayesWave* in the two following rows for a randomly selected injection event. Similarly for the sine-Gaussian glitch injections in Figure 4.3b. The median signal and glitch waveforms in both figures are obtained by taking the median value at each discrete time step from the respective waveform posteriors. The 90% CIs are the range of strain within which the central 90% of the waveform posterior values lies at each time step.

For the sine-Gaussian signal injection, the *BayesWave* signal and glitch models appear to be very similar, that is to say they represent the injected waveform equally well. As discussed in Section 2.3, glitch models have higher dimensionality in detector networks with $\mathcal{I} \geq 2$ and are therefore penalised by the Occam factor resulting in $\ln \mathcal{B}_{S,G}$ favouring the signal model as seen in Figure 4.1.

In contrast, the *BayesWave* signal and glitch model for the sine-Gaussian glitch injections show noticeable differences. Since the injected glitches are independent and dissimilar between detectors, *BayesWave* is unable to construct a signal model that matches the data in all detectors. We can see from Figure 4.2 that the glitch model more accurately represents the injected waveform. This means that the glitch model will have a higher likelihood than the signal model, which then suppresses the effect of Occam penalty, resulting in $\ln \mathcal{B}_{S,G}$ favouring the glitch model as observed in Figure 4.2.



(a) Top row from left to right shows the sine-Gaussian signal injected into the Hanford (H1), Livingston (L1) and Virgo (V1) detectors respectively. Middle row, in the same detector sequence, shows the median *BayesWave* signal waveform along with the 90% credible intervals (CI). Similarly in the bottom row, but for *BayesWave* glitch waveforms.



(b) Top row from left to right shows the independent, coincident sine-Gaussian glitches injected into the Hanford (H1), Livingston (L1) and Virgo (V1) detectors respectively. Middle row, in the same detector sequence, shows the median *BayesWave* signal waveform along with the 90% credible intervals (CI). Similarly in the bottom row, but for *BayesWave* glitch waveforms.

Figure 4.3: Injected waveform and *BayesWave* signal and glitch waveforms in the time domain.

Chapter 5

Binary Black Hole Detection and Waveform Reconstruction

Now that we have successfully gauged the reliability of *BayesWave* in identifying signals and glitches, we can assess the detection confidence of the algorithm for different detector configurations with BBH injections and using Bayes Factor as a figure of merit. In Section 5.1, we present an analysis on the Bayes factor of BBH waveform recovery. This is an extension to that in Section 4.1 such that we now also consider the scaling of the Bayes factor with the number of wavelets. We also quantify *BayesWave*'s performance in waveform recovery with the network overlap in Section 5.2. In Section 5.3, we measure and compare the accuracy of sky localisation between different detector configurations using the areas of 50% and 90% credible intervals as well as the search area. We also discuss the aspects of model-independent waveform moment estimation in Section 5.4.

5.1 Bayes Factor Comparison

By injecting 150 BBH signals into the HL, HLV, and HLKV network data as described in Section 3.4, we study how the log signal-to-glitch Bayes Factor, $\ln \mathcal{B}_{S,G}$, and hence *BayesWave*'s confidence of detection varies between detector networks of three different sizes. For all three detector configurations, we look at how the Bayes factor scales with SNR_{net} (injected SNR) and the number of wavelets, N .

For all the analyses to follow, we will only be using injections that have been identified as inconsistent with Gaussian noise (this can be either a signal or glitch) by *BayesWave*. Injections indicated to be consistent with the Gaussian noise model ($\mathcal{N}$) by *BayesWave* are removed from the data set, as it would be meaningless to evaluate their respective signal and glitch model evidences. In other words, injections with error bars for $\ln \mathcal{B}_{S,\mathcal{N}}$ encompassing values below zero are removed from the dataset. The width of error bars of $\ln \mathcal{B}_{S,\mathcal{N}}$ is given by [22]

$$\Delta[\ln \mathcal{B}_{S,\mathcal{N}}] = \sqrt{\{\Delta[\ln p(d|\mathcal{S})]\}^2 + \{\Delta[\ln p(d|\mathcal{N})]\}^2} \quad (5.1)$$

where $\Delta[\ln p(d|\mathcal{M})]$ is the uncertainty¹ for the log evidence of model $\mathcal{M}$. A total of 15 data points were removed under this constraint and these events were all injected with low SNRs. This leaves us with 135 BBH injections for the analysis.

¹The method for calculating the statistical evidence error in an RJMCMC is given in Reference [22].

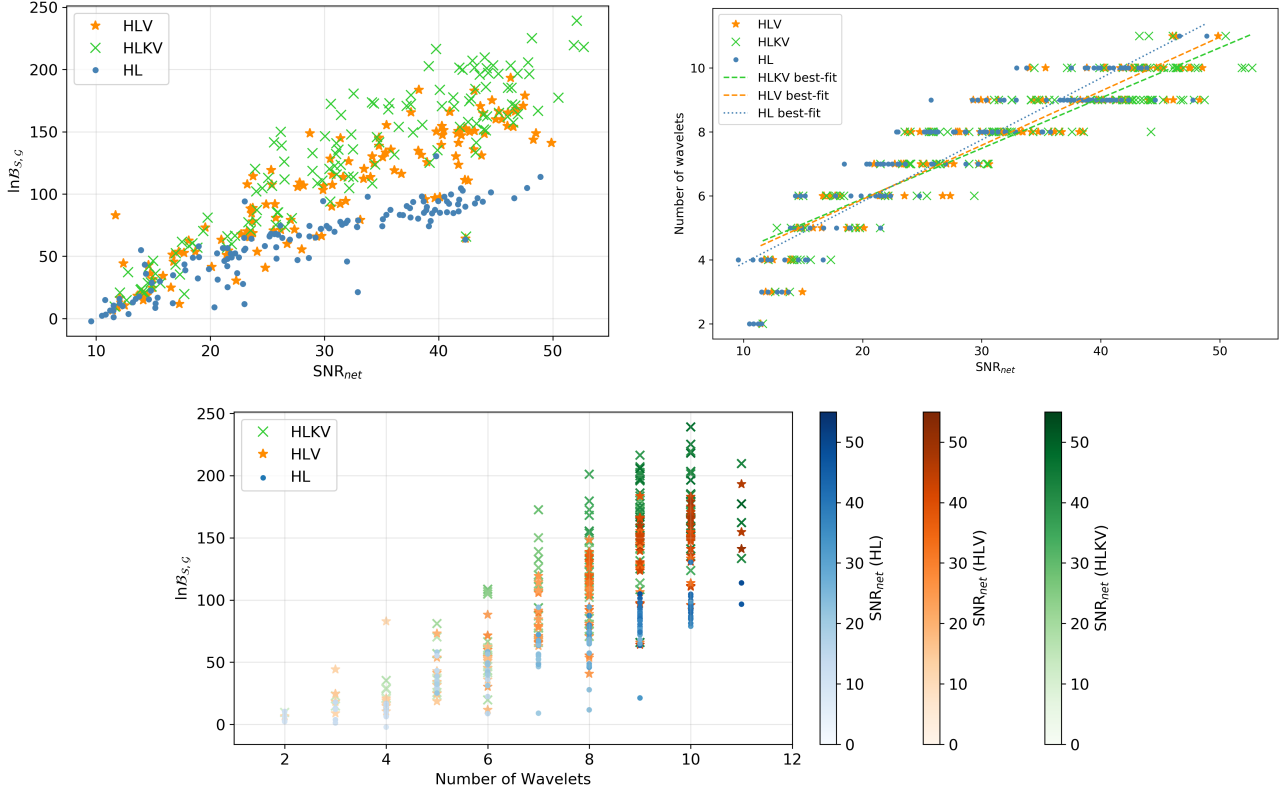


Figure 5.1: (Top left panel) Log signal-to-glitch Bayes Factor $\ln \mathcal{B}_{S,G}$ of BBH injection recoveries versus network signal-to-noise ratio, SNR_{net} . Each data point represents one BBH injection. **(Top right panel)** Median number of wavelets used in signal model reconstruction for each injection, N versus SNR_{net} . Horizontal axes in both top panels are the same as described in Figure 4.1. **(Bottom panel)** $\ln \mathcal{B}_{S,G}$ versus N , and the three colour bars indicates the network SNR of each data point in the corresponding detector network. The blue dot, orange star and green cross symbols represent HL, HLV and HLKV injections respectively.

The top left panel of Figure 5.1 shows $\ln \mathcal{B}_{S,G}$ as a function of SNR_{net} for the HL, HLV and HLKV networks. All three networks show a clear trend of increasing Bayes Factor with increasing network SNR as expected. Our results also show that the HLKV injections have the highest SNR overall, agreeing with Equation 1.9 which indicates that increasing the number of detectors increases SNR_{net} . Furthermore, we can see that injections at comparable SNRs are recovered with higher $\mathcal{B}_{S,G}$ in the HLV network than the HL network. In other words, even after accounting for the increased SNR, we observe further enhancement in detection confidence for an expanded detector network, suggesting that $\ln \mathcal{B}_{S,G}$ is related to the number of detectors in the network, and not just the SNR of the signal as predicted by Equation 2.29.

Unlike the sine-Gaussian wavelet injections in Section 4.1, we should expect to see variations in the number of wavelets used by *BayesWave* to reconstruct BBH waveforms injected at different SNRs. Since the number of wavelets used is itself a parameter of the RJMCMC used by *BayesWave*, the algorithm outputs a posterior for the number of wavelets used in each signal model reconstruction, N . We show the posterior distributions of N for the HL, HLV and HLKV network for one randomly selected injection event in Figure 5.2. Notice that all the posteriors of N have similar shape and the same modal value. However, larger detector networks have distributions skewed more towards larger N . Therefore it is more appropriate to take the median of the posterior as the overall indicative number of wavelets N used in signal model reconstruction for each injected waveform.

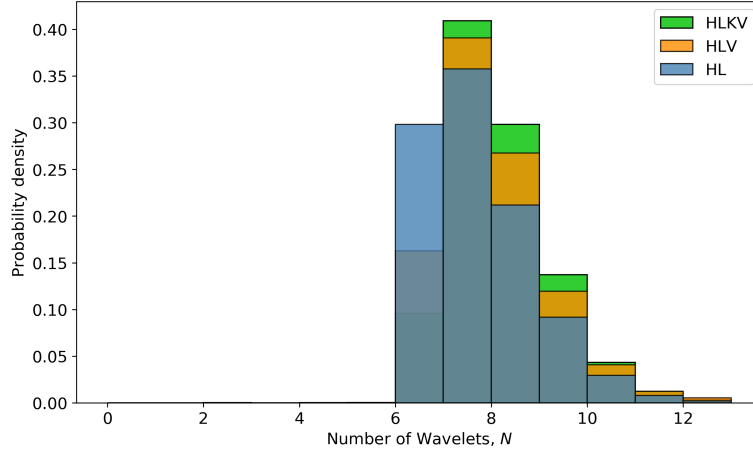


Figure 5.2: Posterior distributions of the number of wavelets used in *BayesWave* signal model reconstruction, N , of a randomly selected BBH injection. The blue bars denote distribution for the HL network, orange for the HLV network and green for the HLKV network.

In the top right panel Figure 5.1, we show a plot of N versus SNR_{net} . We can see that N increases systematically with SNR_{net} in all three detector configurations. This is expected since the detectors are able to pick up more complex features of the waveform at high SNR and has also been empirically validated in [50]. At low SNR, there is a slight deviation from the linear trend described by Equation 2.4 between N and SNR in all three detector networks. This is primarily due to the prior on the number of wavelets. This prior was determined empirically from runs in LIGO data after the O1, and peaks around $N = 3$ [21]. The overall trend of SNR vs. N also appears similar for all three detector configurations. For each network, we fit a least-squares regression line for N in the form of Equation 2.4. The regression lines are indicated as “best-fit” in top right panel of Figure 5.1. The best-fit scaling parameters (η, γ) for the HL, HLV and HLKV networks are given by (2.0, 0.19), (2.5, 0.17) and (2.8, 0.16) respectively up to two significant figures. Previous studies [45, 50] have shown that η and γ vary notably with waveform morphology. For an SNR of 30, the (η, γ) values for all three detector configurations predict there will be 8 wavelets ($N = 8$) used in the waveform reconstruction. Since identical waveforms were injected into all three networks, the results indicate that the number of detectors in the network does not have significant effects on the relationship between SNR and N .

Because Equation 2.29 suggests that $\ln \mathcal{B}_{S,G}$ also scales with the number of wavelets used in the *BayesWave* reconstruction, we also consider whether dimensionalities of signal models (i.e. the number of wavelets) also contribute to the increase in $\mathcal{B}_{S,G}$ for detector configurations of different sizes. We show the plot of $\ln \mathcal{B}_{S,G}$ versus N in the bottom panel of Figure 5.1 along with colour bars indicating SNR_{net} of each data point. For all three detector configurations, $\ln \mathcal{B}_{S,G}$ generally increases with N , as predicted by Equation 2.29. At low SNRs (i.e. $\text{SNR} < 15$), detector networks recover the waveform with less N and low $\ln \mathcal{B}_{S,G}$ because low SNR injections have low amplitude features in the true waveform which can be easily overlooked. The low Bayes Factors suggests that the pipeline is unable to identify a suitable signal model for the true waveform with high confidence. As SNR_{net} increases, N and $\ln \mathcal{B}_{S,G}$ also increase accordingly, signifying that waveforms with signal models of increasing complexity have a higher likelihood to be classified as a signal. Furthermore, it is obvious for $N > 3$ (i.e. high SNR injections) that $\ln \mathcal{B}_{S,G}$ in the HLKV network are generally higher than that of the HL and HLV networks at comparable N . This again emphasizes the point that the Bayes factor is directly related to the number of detectors used in the analysis.

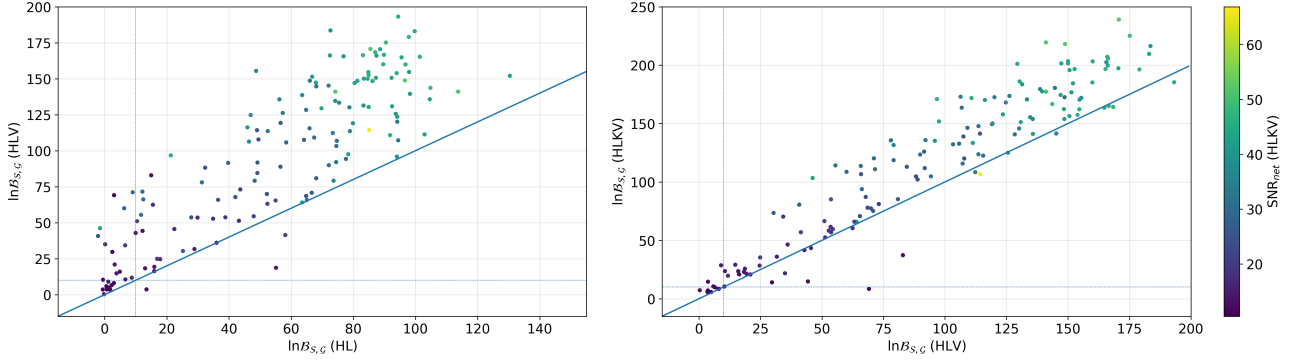


Figure 5.3: (Left panel) $\ln \mathcal{B}_{S,G}$ of the HLV network versus that of the HL network. (Right panel) $\ln \mathcal{B}_{S,G}$ of the HLKV network versus that of the HLV network. The color bar shows the SNR_{net} of the HLKV network for each injection and is applicable to both plots in the bottom panel. The blue lines in bottom panels indicate equal $\mathcal{B}_{S,G}$ in the vertical and horizontal axes.

In summary, these results are consistent with the analytic results discussed Section 2.3.3, viz. $\ln \mathcal{B}_{S,G} \sim \mathcal{O}[\ln \ln \text{SNR}_{\text{net}}]$, where $\ln \mathcal{B}_{S,G}$ and hence detection confidence improves with additional detectors in the network. This conclusion also supports the conjecture from Occam’s Razor that signal models have comparatively lower dimensionalities than glitch models in larger detector networks and are thus more favourable. Heuristically, if coincident identical glitches are unlikely in two detectors, they are even more unlikely in three (or more) detectors. Therefore when coincident identical waveforms are detected across larger networks, it has a higher likelihood of being a signal.

5.1.1 Quantifying Improvements in Bayes Factors

Since the BBH waveform injections for all three detector networks are identical, the three data sets are inherently linked. Hence we can plot $\ln \mathcal{B}_{S,G}$ of two detector configurations against each other to make paired comparison of the Bayes factor (i.e. detection confidence) for detector networks of different sizes. In the left panel of Figure 5.3, we show the plot for $\ln \mathcal{B}_{S,G}$ of the HLV network versus the HL network, and similarly in the right panel for the HLKV network versus the HLV network. The blue lines in bottom panels have gradients of 1 which indicate equal $\mathcal{B}_{S,G}$ in the vertical and horizontal axes. Data points above the line are injections recovered with higher $\mathcal{B}_{S,G}$ from the network in the vertical axis.

We can see that from the left panel of Figure 5.3 that the recovered $\ln \mathcal{B}_{S,G}$ for injections recovered from the HLV network is generally higher than that of the HL network at all SNR, although the difference in $\ln \mathcal{B}_{S,G}$ between the two network becomes increasingly obvious at higher SNR. By Equation 2.30, the improvement in Bayes factor when adding a third detector to the network should be 50%. The best-fit least square, linear regression line for the $\ln \mathcal{B}_{S,G}$ of HLV versus HL has a slope of 1.34 which indicates an average 34% improvement in the Bayes factors with the addition of the Virgo detector. Similarly, we see improvements in $\mathcal{B}_{S,G}$ for the HLKV network compared to the HLV network from the right panel. The slope of the best-fit least square, linear regression line is 1.15 (i.e. 15% improvement), whereas the improvement in Bayes factor predicted by Equation 2.30 from a three- to four-detector network is 33%. The analytic prediction is based on optimal scenarios, so although the empirical improvement in Bayes factor is slightly smaller than the analytic prediction, they still show reasonable agreement.

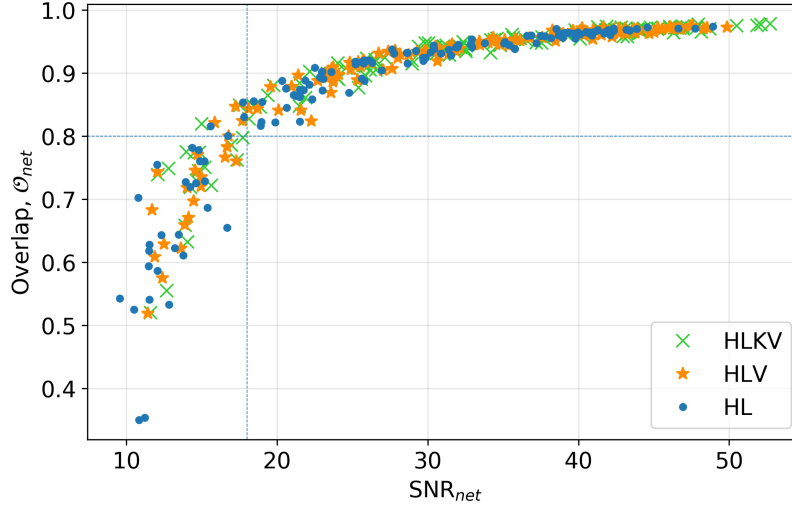


Figure 5.4: Median overlap between the injected and recovered waveform, O_{net} of the HL and HLV network, as a function of SNR_{net} . The horizontal axis is, again, the same as described in Figure 4.1.

Altogether we show by comparing Bayes Factors between the HL, HLV and HLKV networks that expanding detector networks increases detection confidence for signal waveforms. Our empirical results are in agreement with the theoretical scaling of Bayes Factor showed in Equation 2.29.

5.2 Network Overlap and Waveform Reconstruction

In the previous section, we showed that for a set of BBH waveforms, $\ln \mathcal{B}_{S,G}$ increases with a larger number of detectors in the network, meaning with more detectors our confidence in detection is strengthened. In this section, we quantify the accuracy of *BayesWave* in waveform recovery by comparing the overlap (also sometimes called the match) between the injected and recovered waveforms for the HL, HLV and HLKV detector networks. The network overlap, O_{net} is given by Equation 2.32. For the rest of this thesis, any mention of overlap refers to the network overlap.

For each waveform in the posterior distribution, we calculate its overlap with the injected waveform to produce a posterior distribution of overlaps. For the remainder of this work, we use the median overlap from that distribution as the indicative O_{net} for each injection. Figure 5.4 shows the median overlap, O_{net} as a function of network SNR, where O_{net} of all three detector networks show similar positive correlation with their respective network SNR. This observation is consistent with previous results, which show that network overlap scales with SNR [32, 50]. This indicates that waveform reconstruction fidelity is not directly related to the number of detectors in the network.

However as noted earlier, increasing the number of detectors increases the network SNR. By comparing the percentage of waveforms recovered with overlap above a given threshold for all three detector configurations, we show that having an additional detector allows us to better reconstruct the signal waveform. The threshold is arbitrarily defined here to be $O_{net} > 0.8$ and is indicated by the horizontal blue line in Figure 5.4. We found that 80.6% of the injections were recovered with $O_{net} > 0.8$ for the HL network, 85.8% for the HLV network and 86.6% for the HLKV network.

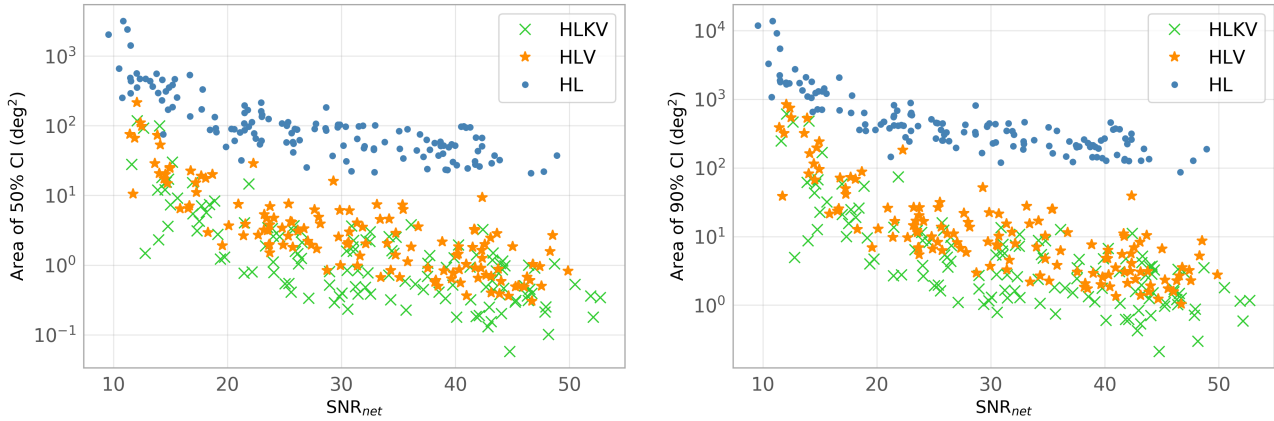


Figure 5.5: The left panel shows the sky area enclosed within the 50% credible interval (CI) in square degrees versus SNR_{net} of the corresponding detector network. Similarly on the right panel for the 90% CI.

While the inclusion of additional detector(s) does not have an extra benefit in the same way it does for the Bayes factor as shown in the previous section, it nonetheless allows us to better reconstruct the signal waveform. However, the improvement is less significant upon the addition of KAGRA because KAGRA is less sensitive compared to Virgo as shown in Figure 3.1 and therefore the increase in SNR is less compared to when Virgo is added to the network. The overall results also show that *BayesWave* is able to reconstruct waveforms reasonably well with all three detector configurations for injections with $\text{SNR}_{\text{net}} > 18$ as indicated by the vertical blue line in Figure 5.4.

5.3 Sky Localisation

Expanding detector networks improves sky localisation of gravitational-wave events. This is shown in results of various studies on coherent network detections e.g. [59] [54] and [53]; see Section 1.5. In this section, we compare the accuracy of *BayesWave* in source localisation between the HL, HLV and HLKV network. We quantify the accuracy sky localisation using two separate measures: (i) sky area enclosed within the 50% and 90% credible intervals (CI) and (ii) search area, $\mathcal{A}$.

For every injection, *BayesWave* outputs posterior distributions for the intrinsic and extrinsic parameters of the waveform. In the first part of this analysis, we combine the posterior distributions of right ascension, α and declination, δ to compute the sky areas enclosed within sky location CIs for source localisation. We denote the combined posterior probability density function (PDF) of sky location as $p_{\text{sky}}(\alpha, \delta)$ and this PDF can be multimodal. The sky areas enclosed within the CIs are obtained by imposing constraints on the parameter domain such that it contains the specified percentage of values in the posterior PDF. For example, the area enclosed within the 90% CI in the context of sky localisation means that it contains the central 90% of the sky coordinates in $p_{\text{sky}}(\alpha, \delta)$. We choose to compare the areas of both the area within the 50% and 90% CIs because although 50% CI provides the source location with less confidence, it imposes stricter constraints on the source location.

In the left panel of Figure 5.5, we show the plot for sky area enclosed within the 50% CI versus network SNR for each injection, and similarly for the the 90% CI on the right panel. For all three detector configurations, we note that the area within the 50% and 90% CIs measured in square degrees

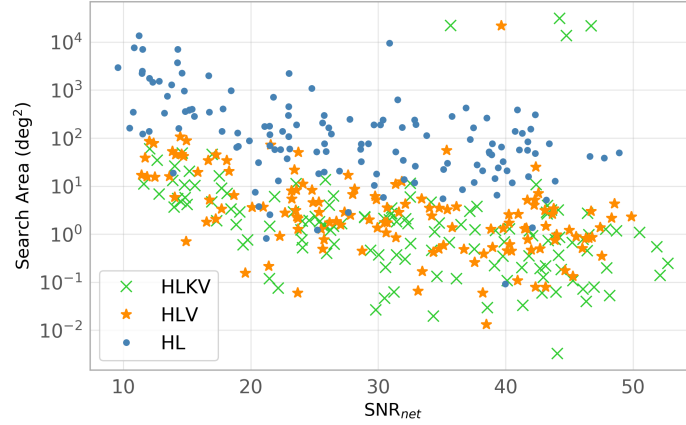


Figure 5.6: Search area, $\mathcal{A}$ (see Equation 5.2) versus SNR_{net} for the HL, HLV and HLKV networks.

(deg²) fundamentally reduces with increasing network SNR due to improved accuracy in arrival time differences [54]. However, both sky areas are generally an order of magnitude smaller for the HLV network compared to the HL network. Upon addition of the KAGRA detector, we observe further reduction in the sky area, but not as drastic as that that between the HL and HLV networks since KAGRA is less sensitive than Virgo. The overall reductions in areas enclosed within both 50% and 90% CIs with increasing number of detectors are attributable to the additional arrival time differences which further constrains the location of each source. From these results, we can conclude that expanding detector network imposes stricter constraints to the source location whilst maintaining the same confidence level.

To visualise the sky areas enclosed within the 50% and 90% CIs, we show two sets of skymaps for two randomly chosen BBH injections of different injected SNR_{net} in Figure 5.7. The left column (from top to bottom) shows the HL, HLV and HLKV skymaps for an injection with SNR_{net} of 14.86, 14.96 and 15.19 in the HL, HLV and HLKV networks respectively. Similarly for the right column for an injection with SNR_{net} of 22.96, 23.43 and 23.55 in the HL, HLV and HLKV networks respectively. The skymaps are two-dimensional histograms with darker regions representing more probable locations of the source. In other words, the coordinates in darker regions appear more frequently in the posterior distributions for sky coordinates: α and δ . The black contour lines (where visible) show the sky areas enclosed within 50% and 90% CIs, and the white star symbol indicates the true location. By comparing the skymaps between different detector configurations, we can conclude as before that sky localisation improves significantly when Virgo is added to the network but only slightly when KAGRA is added. We also observe reduction in areas 50% and 90% CIs in all three detector configurations for the higher SNR injection, consistent with the results in Figure 5.5

Even though the sky areas enclosed within the 50% and 90% CIs provide us with a range of possible values for the true sky location, the estimates are made with no relation to the true location of the source. We can compare the inferred sky location with the true injected location using the search area, $\mathcal{A}$. This metric is the hypothetical sky area observed by a telescope before it correctly points towards the true location of the source. If the true location of the source is (α_t, δ_t) and $p_0 = p_{\text{sky}}(\alpha_t, \delta_t)$, then all points within $\mathcal{A}$ should have $p_{\text{sky}} \geq p_0$. Mathematically, we write [15, 27]

$$\mathcal{A} = \int H[p_{\text{sky}}(\alpha, \delta) - p_0] d\Omega \quad (5.2)$$

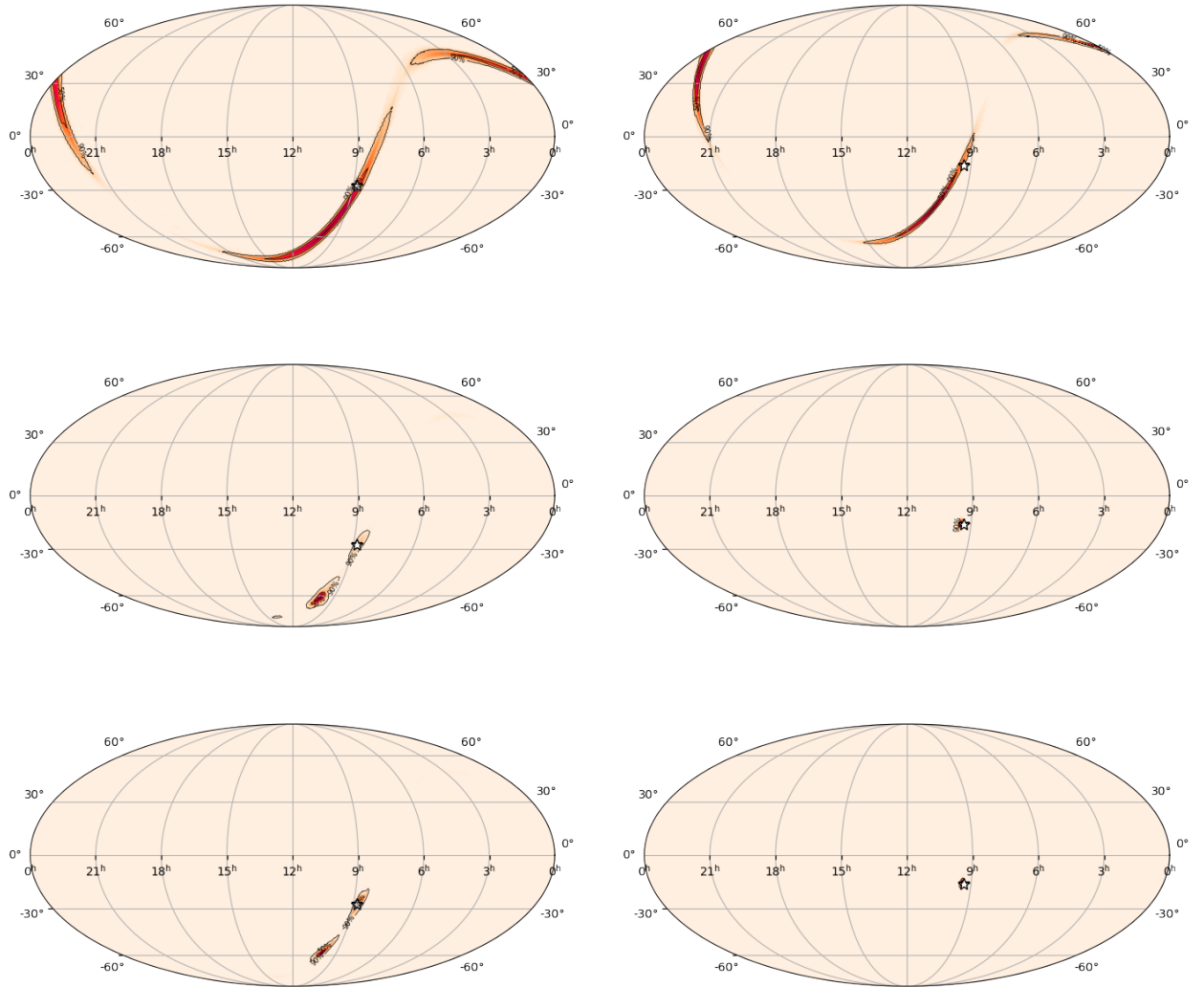


Figure 5.7: Source localisation skymaps for two BBH injection events: **(Left column)** injected with SNR_{net} of 14.86, 14.96 and 15.19 in the HL, HLV and HLKV networks respectively and **(Right column)** injected with SNR_{net} of 22.96, 23.43 and 23.55 in the HL, HLV and HLKV networks respectively. From top to bottom, the rows show the skymaps for the HL, HLV and HLKV networks. Darker regions indicate regions of higher probability in the α and δ posterior distributions, black contour lines (where visible) show the sky area enclosed within the 50% and 90% CIs as labelled, and the white star symbol indicates the true location of the source.

where H is the Heaviside step function and $d\Omega$ is the surface area element on the celestial sphere i.e. $d\Omega = \cos \delta d\delta d\alpha$. Since $\mathcal{A}$ is calculated as a function of the estimated locations relative to the true location of the source, it allows us to gauge the accuracy of the inferred source locations. In Figure 5.6 we plot the $\mathcal{A}$ of all three detector configurations against their respective network SNRs. The search area for HLV network is significantly smaller overall compared to the HL network, whereas the HLKV network show only minor improvements compared to the HLV network. This result is consistent with that shown in Figure 5.5, indicating sky localisation improves remarkably when a detector of high-sensitivity is added to the network. If a less sensitive detector is added, the improvements are small but not negligible.

5.4 Waveform Moments

Since generic burst signals do not have robust astrophysical models, algorithms like *BayesWave* do not directly output information about the signal source. We can however characterize the signal, independent of an astrophysical model, using the so-called waveform moments. These waveform moments can in turn be used to constrain some of the source properties such as the characteristic mass [16]. In this final section, we present an analysis on *BayesWave*'s performance in estimating model-independent waveform moments from the HL, HLV and HLKV networks.

First-order central moments of interest are the central time (t_0) and central frequency (f_0), and second-order central moments are the root-mean-square duration (Δt) and the bandwidth (Δf):

$$t_0 = \int_{-\infty}^{\infty} dt \rho_{\text{TD}}(t)t \quad (5.3)$$

$$f_0 = \int_0^{\infty} df \rho_{\text{FD}}(f)f \quad (5.4)$$

$$(\Delta t)^2 = \int_{-\infty}^{\infty} dt \rho_{\text{TD}}(t)(t - t_0)^2 \quad (5.5)$$

$$(\Delta f)^2 = \int_0^{\infty} df \rho_{\text{FD}}(f)(f - f_0)^2. \quad (5.6)$$

Here ρ is the effective normalised distribution of signal energy, and can be computed in the time or frequency domain:

$$\rho_{\text{TD}}(t) = \frac{h(t)^2}{h_{\text{rss}}^2} \quad (5.7)$$

$$\rho_{\text{FD}}(f) = \frac{2|\tilde{h}(f)|^2}{h_{\text{rss}}^2} \quad (5.8)$$

where $h_{\text{rss}}^2 = \int (h_+^2 + h_\times^2) dt$ is the root-sum-squared strain amplitude. The subscripts TD and FD denote time domain and frequency domain respectively.

BayesWave produces a posterior distribution of the waveform present in detector data, from which we can obtain posterior distributions on the waveform moments. We then follow [15] by defining the absolute error of a waveform moment, x as

$$e_x = |x^{(e)} - x^{(r)}| \quad (5.9)$$

where $x^{(e)}$ is the point estimate (posterior median) of x recovered by *BayesWave* and $x^{(r)}$ is the true value of x for injected waveform. For the first-order moments t_0 and f_0 , we compare the accuracy of estimation using the ratio between the absolute error and the true values of corresponding second-order moments i.e. $e_{t_0}/\Delta t^{(r)}$ and $e_{f_0}/\Delta f^{(r)}$ since the statistical error of first-order moment estimation scales with the true values of the second-order moments [15]. For the second-order moment Δt and Δf , on the other hand, we quantify the estimation accuracies using relative errors

$$\eta_x = \frac{e_x}{x^{(r)}}. \quad (5.10)$$

Since *BayesWave* estimates the waveform moments individually for each detector, we infer the accuracy of the algorithm in estimating waveform moments by comparing the injected and recovered values from the Hanford detector data, since the Hanford detector is present in all three the HL, HLV and HLKV networks. The same inference can be drawn if we were to make the comparison using waveform moment estimations for the Livingston data, but not for the Virgo or KAGRA detectors since they not present in all three networks of interest.

We show the comparison of waveform moment estimations between the three detector configurations in Figure 5.8. The left column from top to bottom show $e_{t_0}/\Delta t^{(r)}$, $\eta_{\Delta t}$, $e_{f_0}/\Delta f^{(r)}$ and $\eta_{\Delta f}$ versus the network SNR of the corresponding detector network. As expected, we see overall improvements in waveform moment estimation with increasing SNR and all three detector networks follow similar trends. Since *BayesWave* is able to identify low amplitude details of the waveform at high SNR, it is able to reconstruct more waveform models with high overlaps with the true waveform (see results in Section 5.2) and hence able to make more accurate estimations of the waveform moments. We also note from the comparison of waveform overlaps that *BayesWave* is able to recover waveforms equally well with the HL, HLV and HLKV networks. Hence, we see from the plots in the left column of Figure 5.8 that the relative errors for all waveform moment estimations are comparable between all three detector networks.

To compare the overall accuracies of *BayesWave* in estimating different waveform moments, we also show the corresponding cumulative distribution functions (CDFs) for each quantity on the logarithmic scale in the right column of Figure 5.8. The CDFs show that the relative errors for second-order moment estimations are larger compared the first-order moments. Since second-order moments (Δt , Δf) are computed from the first-order moments (t_0 , f_0) as per Equation 5.5 and 5.6, the error propagates forward from lower-order moment estimation and are hence subjected to larger statistical errors. There are some noticeable differences between the CDFs of different detector configurations which might signify improvements in moment estimations. However, the CDFs are constructed with the same dataset as for the relative errors versus SNR plots, so it still applies that *BayesWave* can estimate the moment equally well with all three detector configurations. In other words, any variations between the CDFs are purely statistical.

From this analysis, we conclude that waveform moment estimation improves with network SNR for all three detector configurations: HL, HLV and HLKV, but there are no significant improvements in waveform moment estimations with expanded detector networks. We also noted that higher-order moments are estimated with reduced accuracy due to error carried forward from lower-order moment estimations.

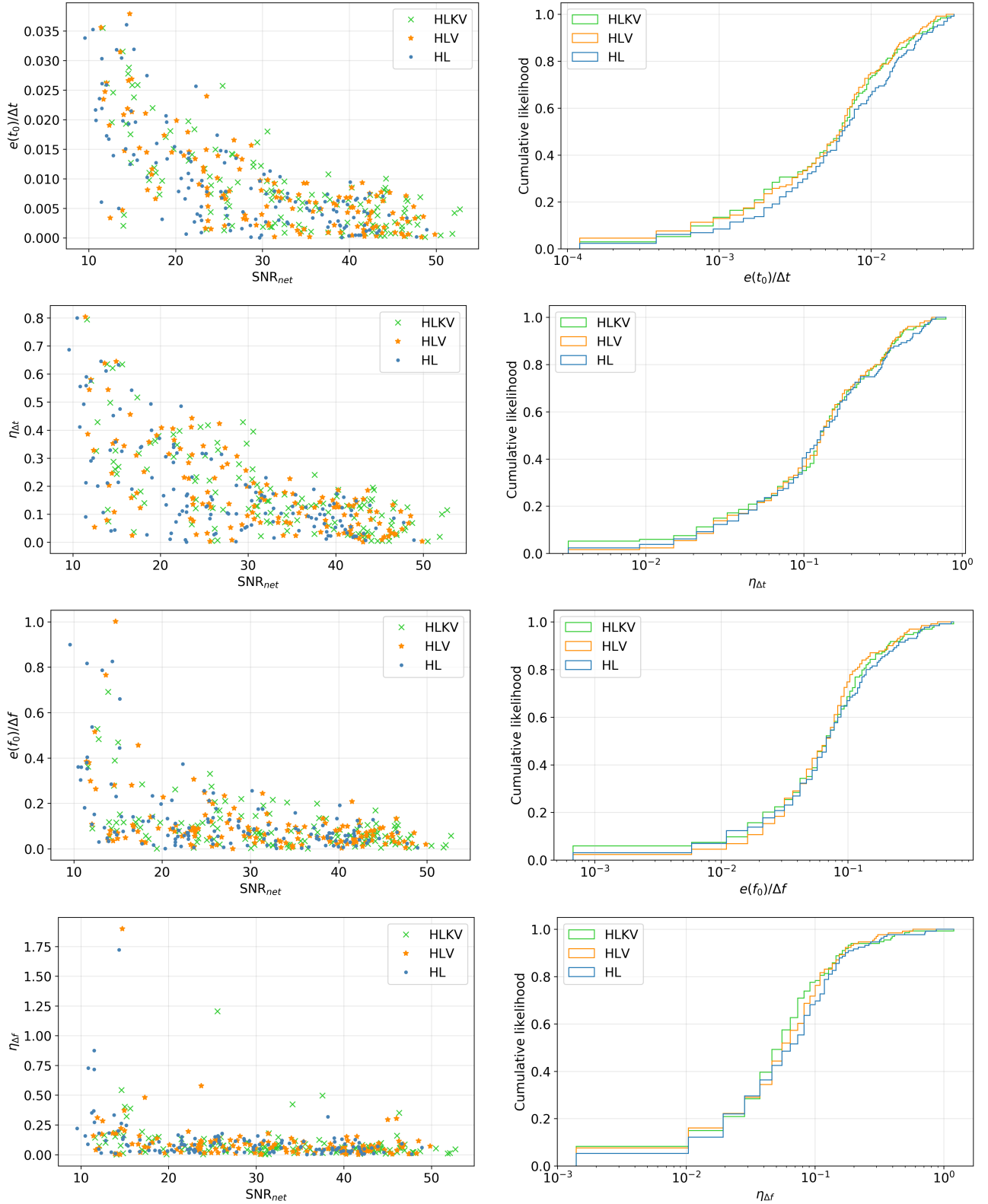


Figure 5.8: (Left column) Relative moment estimation errors versus network SNR of the corresponding detector network. From top to bottom: ratio between absolute errors of central time and signal duration ($e_{t_0}/\Delta t^{(r)}$), relative errors of signal duration estimation ($\eta_{\Delta t}$), ratio between absolute errors of central frequency and signal frequency bandwidth $e_{f_0}/\Delta f^{(r)}$ and relative errors of bandwidth estimation $\eta_{\Delta f}$. **(Right column)** Cumulative distribution functions (CDF) corresponding to the relative error in the left column. Colours indicate CDFs for relative errors of the HL (blue), HLV (orange) and HLKV (green) networks.

Chapter 6

Conclusions and Future Work

In this work, we compare the performance of *BayesWave* in recovering gravitational-wave signals from detector networks of different sizes. We first derived an analytic scaling for the Bayes factor between the signal and glitch models in Section 2.3.3, and performed consistency tests on the *BayesWave* algorithm with sine-Gaussian signal and glitch injections in Chapter 4. We show that *BayesWave* correctly identifies signals and glitches from detectors of different sizes, and also preliminarily validated the analytic scaling of the Bayes factor. We then inject the same sets of simulated BBH signals of fixed masses at different SNRs into simulated O4 detector data of the HL, HLV and HLKV network to evaluate *BayesWave*'s performance in signal identification and waveform recovery.

Previous work [45] demonstrated that *BayesWave* is unique amongst unmodelled gravitational wave burst searches in that the complexity of the signal in time-frequency space plays a crucial role in the detection statistic, rather than just the signal's strength. Our results in Section 5.1 are consistent with [45] and the theoretical scaling $\ln \mathcal{B}_{S,G} \sim \mathcal{O}[\mathcal{I}N \ln \text{SNR}]$, which indicates that waveforms with more complex structures need more wavelets, N to accurately reconstruct the waveform and hence have higher $\ln \mathcal{B}_{S,G}$. Additionally, we demonstrate another novel feature of the *BayesWave* algorithm: the detection statistic is also influenced by the number of detectors, $\mathcal{I}$. For signals of similar injected SNR (SNR_{net}), events analysed using larger detector networks have higher $\ln \mathcal{B}_{S,G}$ indicating detection confidence increases more than we would expect purely from the increase in SNR_{net} . In Section 5.2, we look at the network overlap, $\mathcal{O}_{\text{net}}$, between the injected and recovered waveforms, and find that it increases with SNR_{net} . Since larger detector networks can detect signals at higher SNR, they are able to pick up more details of the true waveform. Thus *BayesWave* can reconstruct the waveforms more accurately with additional detectors. We also show that search area $\mathcal{A}$, and the sky areas enclosed within the 50% and 90% credible intervals (CI) decrease with increasing SNR_{net} in Section 5.3, suggesting that increasing $\mathcal{I}$ at different geographical locations can improve sky localisation. Finally in Section 5.4, we quantify *BayesWaves*' signal characterisation accuracy by comparing the relative errors in waveform moment estimation. Even though the accuracy of moment estimations do not show significant improvement with larger detector networks, it generally improves with SNR_{net} .

In summary, *BayesWave* shows significant improvements in terms of signal identification and waveform recovery when working with a larger detector network. This promising result suggests that with more detectors joining the global network in the future, we will be able to reconstruct generic gravitational-wave burst signals more accurately using *BayesWave* and make detections with higher Bayes factor, hence with higher confidence. Prospective work along the lines of this work include injecting different waveform morphologies to compare detection confidence between detector networks of different sizes. We also recommend looking into quantifying and comparing the outcomes of *BayesWave* in recovering simulated signals from more realistic detector noise (i.e. in the presence of glitches) between different detector configurations.

Appendix A

Model evidences for Bayes factor scaling

In Section 2.3.3, we use the logarithmic signal and glitch model evidences given by Equations 2.20 and 2.21 to obtain a general scaling relation for the logarithmic signal-to-glitch Bayes Factor. Following methods described in Reference [45], we reproduce the derivations of Equations 2.20 and 2.21 from the Laplace approximation of model evidences (Equation 2.14, shown below)

$$p(\vec{s}|\mathcal{M}) \simeq p(\vec{s}|\vec{\theta}_{\text{MAP}}, \mathcal{M})p(\vec{\theta}_{\text{MAP}}|\mathcal{M})(2\pi)^{D/2}\sqrt{\det C}.$$

Recall that model parameters can be classified into two separate groups: intrinsic and extrinsic, so we first consider the intrinsic parameters, λ . The first factor in Equation 2.14 is the MAP likelihood, $p(\vec{s}|\vec{\theta}_{\text{MAP}}, \mathcal{M})$ and the expectation value of intrinsic parameter logarithmic likelihood is given by [56]

$$\ln p(\lambda_{\text{MAP}}|\mathcal{M}) \sim \frac{\text{SNR}^2}{2} - \frac{D}{2} \quad (\text{A.1})$$

For the MAP prior, $p(\vec{\theta}_{\text{MAP}}|\mathcal{M})$, since *BayesWave* uses uniform priors for all intrinsic parameters except for the amplitude parameter, A , we proceed to use the following relation for simplicity [22]

$$p(\lambda_{\text{MAP}}|\mathcal{M}) = \left(\frac{1}{V_\lambda}\right)^N \quad (\text{A.2})$$

where we neglect the parameter-dependence of the amplitude prior (see Equation 3.3). The power N indicates that the total intrinsic parameter space volume is the product of N wavelets, each with equal parameter space volume, V_λ . To obtain the determinant of the full intrinsic parameter covariance matrix, C_λ , we first consider the determinant of the single wavelet FIM in Equation 2.16

$$\det \Gamma = \frac{\pi^2 \text{SNR}^{10}}{2Q^2}. \quad (\text{A.3})$$

Assuming that the parameter spaces of the N wavelets have overlap minimally, the full FIM, Γ_λ for N wavelets will be a block diagonal with its determinant given by

$$\det \Gamma_\lambda = \prod_{n=1}^N \frac{\pi^2 \text{SNR}_n^{10}}{2Q_n^2}. \quad (\text{A.4})$$

where the subscripts n denotes an individual wavelet from the set of N wavelets. It then follows that

$$\sqrt{\det C_\lambda} \approx \prod_{n=1}^N \frac{\sqrt{2}Q_n}{\pi \text{SNR}_n^5}. \quad (\text{A.5})$$

since $\det C = 1/\det \Gamma$. We then set the dimension of the model to $D = 5N$ since there are N wavelets each with 5 intrinsic parameters, and define $\bar{Q}_n = (2\pi)^{5/2} \frac{\sqrt{2}Q_n}{\pi}$ such that

$$\ln \left[(2\pi)^{5N/2} \sqrt{\det C} \right] = \sum_{n=1}^N \ln \frac{\bar{Q}_n}{\text{SNR}_n^5} \quad (\text{A.6})$$

Glitch models only consists of intrinsic parameters, so the logarithmic glitch model evidence for a single detector i , $\ln p(\vec{s} | \mathcal{G}_i)$ is obtained by substituting Equations A.1, A.2 and A.6 into the logarithmic form of Equation 2.14

$$\ln p(\vec{s} | \mathcal{G}_i) \simeq \frac{\text{SNR}_i^2}{2} - \frac{5N_i^{\mathcal{G}}}{2} - N_i^{\mathcal{G}} \ln(V_\lambda) + \sum_{n=1}^{N_i^{\mathcal{G}}} \ln \left(\frac{\bar{Q}_n}{\text{SNR}_{i,n}^5} \right) \quad (\text{A.7})$$

Since glitches are independent in each detector, the overall logarithmic glitch model evidence (Equation 2.21, shown below) is a sum over $\ln p(d | \mathcal{G}_i)$ of individual detectors i

$$\ln p(\vec{s} | \mathcal{G}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \sum_i^{\mathcal{I}} \left[\frac{5N_i^{\mathcal{G}}}{2} + N_i^{\mathcal{G}} \ln(V_\lambda) - \sum_{n=1}^{N_i^{\mathcal{G}}} \ln \left(\frac{\bar{Q}_n}{\text{SNR}_{i,n}^5} \right) \right]$$

where $\mathcal{I}$ is the total number of detectors in the network, $N_i^{\mathcal{G}}$ is the number of wavelets used to fit the glitch model in detector i , $\text{SNR}_{i,n}$ is the SNR of wavelet n in detector i and $\text{SNR}_{\text{net}}^2$ came from the quadrature sum of SNRs of each detector as per Equation 1.9.

Signal models, on the other hand, only require one waveform model to fit all the detector data so the sum over i is not necessary. However, additional terms $D_\Omega/2$ and $\ln(\sqrt{\det C_\Omega}/V_\Omega)$ are required to account for extrinsic parameter dimensions and parameter space volumes respectively. Altogether this gives the logarithmic signal model evidence (Equation 2.20, shown below)

$$\ln p(\vec{s} | \mathcal{S}) \simeq \frac{\text{SNR}_{\text{net}}^2}{2} - \frac{5N^{\mathcal{S}}}{2} - N^{\mathcal{S}} \ln(V_\lambda) + \sum_{n=1}^{N^{\mathcal{S}}} \ln \left(\frac{\bar{Q}_n}{\text{SNR}_{\text{net},n}^5} \right) + \frac{D_\Omega}{2} + \ln \frac{\sqrt{\det C_\Omega}}{V_\Omega}.$$

where $N^{\mathcal{S}}$ is the number of wavelets used to fit the signal model. The prior volume of extrinsic parameters is $V_\Omega = 4\pi^2$ and since there are four extrinsic parameter, we can set $D_\Omega = 4$.

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